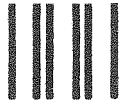
HYPERTIES™

HYPERTEXT SOFTWARE FOR THE IBM-PC®

COGNETICS® CORPORATION



NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 100 PRINCETON JCT., NJ

POSTAGE WILL BE PAID BY ADDRESSEE

The Hyperties[tm] Project
Cognetics Corporation
55 Princeton-Hightstown Rd.
Princeton Jct., NJ 08550

Product Serial Number _____

Hyperties™ User Feedback

We would appreciate your comments about Hyperties™ and this manual. Please let us know about any errors or bugs you uncover, or features you would like added.

NAME _____

TITLE **COMPANY** _____

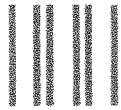
DIVISION **PHONE** _____

ADDRESS _____

ADDRESS _____

CITY **STATE** **ZIP** _____

(Fold, tape together, and mail -- no stamp needed)



NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

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FIRST CLASS PERMIT NO. 100 PRINCETON JCT., NJ

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The Hyperties™ Project
Cognetics Corporation
55 Princeton-Hightstown Rd.
Princeton Jct., NJ 08550

Product Serial Number _____

Hyperties[tm] User Registration

Please return this user registration form immediately so we have your correct mailing address and can send you updates.

NAME _____

TITLE _____ **COMPANY** _____

DIVISION _____ **PHONE** _____

ADDRESS _____

ADDRESS _____

CITY _____ **STATE** _____ **ZIP** _____

What Computer(s) will you be using?

[ ] IBM PC [ ] IBM XT [ ] IBM AT [ ] IBM PS/2 Model _____

[ ] IBM Compatible Model _____ MEMORY SIZE _____K

What type(s) of disk(s)?

[ ] Hard Disk [ ] 5-1/4" Floppy (360K) [ ] 5-1/4" Floppy (1.2 Meg)

[ ] 3-1/2" Floppy (720K) [ ] 3-1/2" Floppy (1.4 Meg)

What type of monitor?

[ ] Monochrome [ ] Hercules [ ] CGA Color [ ] EGA Color

[ ] VGA Color [ ] Other _____

Planned Use of Hyperties[tm]?

_____

_____

_____

_____

Hyperties™

Quick Reference

Starting the Browser

To start *The Browser* type:

BROWSER *directory*

where:

directory is the path name of the directory with the Hyperties™ data base.

See the manual for information on activating the mouse for input.

Getting Around a Database

Use the arrow keys to move the *selector bar* to a highlighted term and press enter.

With a mouse, place the bar on the desired highlighted term and click.

Browser Hot Keys

PgUp	Previous Page
PgDn	Next Page
Home	First Page of Article
End	Last Page of Article

(continued on next side)

Search String	Finds Articles Containing
word	the specified word
word word2	word followed by word2
word&word2	both word and word2
word|word2	either word or word2 or both

The database must have signature files (SIG.S and SIG.P) created by the UPDATE utility. Otherwise search will not be activated on the index page.

History

Select EXTRA, then HISTORY. Shows all articles visited.

Contents

Select EXTRA then TABLE OF CONTENTS. Only active if one was included in the database.

Starting the Author

Create a directory to contain the database if needed. To start *The Author* type:

AUTHOR *directory*

Make certain *The Browser* is in the same directory or accessible with a PATH command if you want to switch between *The Author* and *The Browser*.

Commands

Use selector bar, press ENTER. If bar is in a window, press Esc to place it on command line.

Browser Hot Keys (Continued)

Esc	Back one article (Return to ...)
<CTRL>-E	EXTRA (Index, History & Search)
<CTRL>-T	Top (jump back to first article read)
<CTRL>-Q	QUIT
<CTRL>-P	Print Current Article
?	Display Help Screen

Press any letter to jump directly to highlighted terms beginning with that letter. Press several times if necessary.

Index

To get to the index, select EXTRA from any article. Articles visited have a * next to name.

You may select any article from the index. Use NEXT PAGE and BACK PAGE to move around the index. Select TURN TO, then a letter, to turn directly to the index page containing that letter.

Select SEARCH or TABLE OF CONTENTS to get to those functions.

Search

Searches articles for occurences of a specific word. Select EXTRA then SEARCH to get to the search facility.

Select the SEARCH STRING to set the string you want to search for.

(continued)

Search String	Finds Articles Containing
word	the specified word
word word2	word followed by word2
word&word2	both word and word2
word|word2	either word or word2 or both

The database must have signature files (SIG.S and SIG.P) created by the UPDATE utility. Otherwise search will not be activated on the index page.

History

Select EXTRA, then HISTORY. Shows all articles visited.

Contents

Select EXTRA then TABLE OF CONTENTS. Only active if one was included in the database.

Starting the Author

Create a directory to contain the database if needed. To start *The Author* type:

AUTHOR *directory*

Make certain *The Browser* is in the same directory or accessible with a PATH command if you want to switch between *The Author* and *The Browser*.

Commands

Use selector bar, press ENTER. If bar is in a window, press Esc to place it on command line.

You may execute any command on the screen by holding down the Alt key and pressing the first letter. Example: **Alt-I = INDEX, Alt-E = EDIT.**

Special Articles

You may create any of the following special articles:

INTRODUCTION
TABLE OF CONTENTS
PREFACE

Embedded Format Commands

@S1 @S2	set spacing
@L*n* **@L+***n* **@L-***n*	left margin (*n*=position)
@R*n* **@R+***n* **@R-***n*	right margin
@M*n* **@M+***n* **@M-***n*	both margin
@T	table mode (toggle on/off)
@P*n* *n* optional	new paragraph (indent *n*)
@N	new page
@A=*xxxxx*	include article *xxxxx* in this
@G=*xxxxx*	insert graphic *xxxxx* here
@B=*xxxxx*	use background graphic *xxxxx*
@V=*xxxxx-yyyyy*	display videodisc *xxxxx-yyyyy*

Editor Commands

Arrows, Ins, Del, Backspace key work as expected.
PgUp PgDn scroll up/down one page.
Home, End go to top/bottom of text.
<CTRL>-Home, <CTRL>-End start/end of line.
<CTRL>-left/right arrow jump one word.

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Princeton Jct. NJ 08550 ▪ (609) 799-5005

HYPERTIES AUTHOR'S GUIDE

Hyperties[tm] License Agreement

Important: Please read this license agreement before opening the envelope containing the disks. By opening the envelope, you demonstrate your agreement to the terms of the license. If you do not wish to follow the terms of the license agreement, you may return the software and receive a full refund as long as you have not opened the envelope with the disks.

INTRODUCTION

This license agreement is for Hyperties[tm] version 2.3. Hyperties[tm] is copyrighted by Cognetics Corporation. You may only use Hyperties[tm] in the manner set forth in this agreement.

This license gives you certain limited rights to use Hyperties[tm] version 2.3. You do not become the owner of Hyperties[tm] and Cognetics Corporation retains title to all software, copies of the software and printed materials supplied to you now or in the future.

PERMITTED USES

You may use the *Author Tool* on a single computer only. You may make up to three (3) back-up copies of the Author Tool for archival purposes only. You must reproduce all information on the disk labels (including the copyright notice) on your back-up disks.

You may transfer the Author Tool to a hard disk on the single computer.

You may distribute copies of the *Browser* with data bases you create providing you do not sell such copies for profit. You may distribute up to twenty (20) copies of each data base you create without securing an additional license from Cognetics Corporation.

You agree to use reasonable care to protect Hyperties[tm] from unauthorized use.

USES NOT PERMITTED

You may not make copies of the software or associated printed materials except as permitted above.

You may not sell the software, copies of the Browser, decompile, disassemble, or reverse engineer the software. If you want to do any of these things, contact Cognetics Corporation to arrange for an extended license.

You may not remove or obscure the copyright or trademark notices.

WARRANTY

We have made diligent efforts to produce error-free code. It is a mathematical impossibility to prove that a program is error-free. Therefore, we do not guarantee the performance of Hyperties[tm]. It is your responsibility to test the databases you create before you put them to use. If you are dissatisfied with the performance of the programs let us know within a reasonable time. We will, at our option, correct the problem or refund your money. Our liability is limited to the price you paid for the programs. We explicitly disclaim liability for lost profits or other damages.

GENERAL

This agreement represents the entire understanding between Cognetics Corporation and you. It may be modified only in writing by an authorized Cognetics officer.

Background Graphics

Select the **Background Graphic** option on the menu when you want to view different colors of text on a particular background graphic. *The color of text can change radically when it is displayed over a background graphic*, so this menu option is helpful in deciding what colors will work right on specific backgrounds.

When "Background Graphic" is selected, a window appears with a list of available graphics, plus two additional options: NONE and NEW DIRECTORY. To choose a graphic, highlight its name with the selector bar and press return. (If there are more than 28 graphic files in the directory, you can view more by selecting the down-pointing arrows at the bottom of the window.) Choose NONE if you want to clear the background graphic, and NEW DIRECTORY if you want to specify a different directory containing graphics.

When a background graphic is chosen, the menu in Figure A.2 is overlaid on the chosen background graphic. In graphics modes, the selector bar color is determined automatically by *The Browser*, so you cannot specify it. Also, the screen border and screen background colors may not be specified.

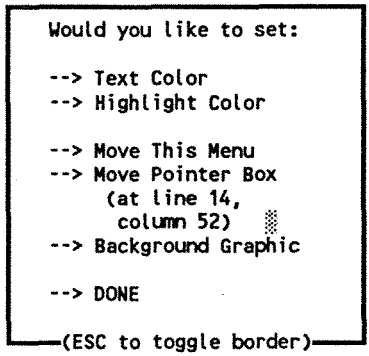


Figure A.2
The Main Menu (graphics mode)

Since one of the main purposes of showing a background graphic in the Palette program is to check out text colors, you may "turn off" the border surrounding the main menu, causing the text of the menu to directly overlay the graphic. To do this, you use the **ESC** key.

Also, there are two additional menu options available: since the *position* of text on a background may influence its color, the menu options **Move This Menu** and **Move Pointer Box** are provided. Move the menu to see how text will look in different places, and move the shaded box (▒) to determine the coordinates of various places on the screen. (This will help when you are formatting the text of articles which will use this graphic as a background.)

When You're Done

When you select DONE from the main menu, the menu in Figure A.3 appears. Again, select an option by moving the selector bar with the arrow keys and pressing ENTER.

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Chapter 1
Introducing Hyperties

This chapter introduces you to the basic concepts of Hyperties™. For information on how to install Hyperties™ on your computer, see the chapter titled *Installing Hyperties™*.

Introducing Hyperties™

Hyperties™ is a powerful, yet simple, hypertext software tool for organizing and presenting information. As a Hyperties™ author, you will be able to create databases consisting of *articles* and *graphics*. Without the need for programming, you will be able to link these articles and graphics together so your readers will be able to browse through them, following paths of interest. Hyperties™ databases are sometimes thought of as electronic books or *hyperbooks* because they are organized like an encyclopedia with articles and cross-references.

Hyperties™ can be used for a wide variety of applications, including:

- On-line encyclopedias
- On-line documentation
- Instruction and dynamic glossaries
- Reference manuals
- Corporate policy manuals
- Summaries of products and services
- Employee orientation
- Regulations and procedures
- Museum exhibits

Development

Hyperties™ was developed at the University of Maryland's Human-Computer Interaction Laboratory under the leadership of Professor Ben Shneiderman and is the product of five years of research. It is distributed by Cognetics Corporation, Princeton

Junction, New Jersey, which is continuing its development in cooperation with the University of Maryland.

Hyperties™ continues to be the focus of an active research program at the University of Maryland and the product is being continually revised and improved. When you mail-in your registration card, you will automatically be registered as a member of the Hyperties™ User's Group (HUG). We look forward to your input and hope you will play an active role in shaping the future development of Hyperties™.

Hyperbooks

Hyperties™ was originally known as *TIES* - The Interactive Encyclopedia System. It is designed to create databases which are "electronic books" or *hyperbooks*. Each database consists of elements which are similar to those in a conventional printed book:

- A Cover
- A Preface
- A Table of Contents
- Articles
- Illustrations
- An Index

You do not need to think of a Hyperties™ database as an electronic book or hyperbook and in some cases the database may be best thought of differently. However, you will often find the idea of a hyperbook a useful one as you design your databases.

The Hyperties Tour

Because the easiest way to become acquainted with Hyperties™ is to try it, we have provided a demonstration database: The Hyperties™ Tour. When you install Hyperties™ on your computer, the Hyperties™ Tour will automatically be installed, as well. Chapter 2 -- *The Browser* will "walk you" through the tour. We recommend that you explore this tour at your earliest convenience.

Authoring and Browsing

Hyperties™ consists of two major programs: *The Author* and *The Browser.*

The Author is the program which you use to create a Hyperties™ database. Using the authoring tool is simple -- like using a familiar word processor. In fact, you may use your favorite word processor to prepare articles for Hyperties™ and "import" them into the database. (You may also "export" articles into your own word processor). When you create a database, you provide Hyperties™ with the text of the articles and the *links* to other articles. Hyperties™ automatically ties the articles and illustrations together into a unified database and constructs an index to the entire database.

The Browser is the program used by your readers to access the Hyperties™ database of articles and graphics you prepare. (*Note: When you explore the Hyperties™ Tour you are using The Browser.*) Using *The Browser* is extremely easy and requires virtually no training. Your readers can access complete articles, definitions of important terms, graphics and cross references by using only three keys: the left arrow key ←, the right arrow key →, and the ENTER key. Hyperties™ also supports the use of a mouse for browsing. See Chapter 9 for details.

If your computer is equipped with a touchscreen, your readers can use it to browse through the database without the use of a keyboard at all. (*Note: A special version of Hyperties™ 2.3 supports the Microtouch™ and IBM Infowindow™ touch screens. This special version also supports videodisc on the IBM Infowindow. Please contact Cognetics Corporation for information).*

Ease of Use

In designing Hyperties™, ease of use was an extremely important design goal. *The Browser* has been designed so that the reader does not need to learn any commands to use it. Every article and command is accessible in the same way -- by selecting a highlighted word or phrase.

As far as possible, the software was designed to prevent errors from occurring or, if they do occur, handling them gracefully. For example, if the author includes a graphic in the database which cannot be displayed on the reader's computer, *The Browser* will construct a page of text, explaining its inability to display the graphic. The program will not "bomb" nor will the reader have to take any action. This philosophy is intended to turn an error condition into a minor inconvenience.

The Author program was also designed to be easy for the non-technical user. It is our belief that in hypertext, the author and user's attention should be focused on the content of the database -- not on technical details.

However, we also recognize that as authors and readers become skilled in using Hyperties™, they will want shortcuts. We have therefore designed-in a number of hidden features for the "power user." Neither the author nor the reader needs ever to learn them but they are available if desired.

Links

The power of Hyperties™ is your ability to create *links* which tie articles together. A link is a cross reference, an indication that more information on a topic is available.

Links permit your reader to browse through the database, following paths of interest. Hyperties™ keeps track of the reader's path and allows the reader to retrace it.

The links in Hyperties™ are words or phrases in the text which are **highlighted** when displayed. Links are always displayed on the screen; the reader does not need to take any action to uncover them. Because links are actually part of the text, the process of following links is a natural one.

Brief Descriptions

Each link in Hyperties™ may be "attached" to a *brief description*. When the reader selects the link, (s)he sees the brief description. If more information on the topic is available (in the form of a full article), the reader may choose to see it.

Hyperties™ has three options for display of the brief description. You may choose the option desired by setting a line in the TIES-.SYS file as described in Chapter 9.

The standard option is to *always* display the brief description. If there isn't one, only the name of the article is displayed.

The second option, which is referred to as *present* is to display a brief description if it exists; if one doesn't exist, go directly to the article attached to the link.

The third option, known as *never*, ignores all brief descriptions. This is particularly useful in exhibit situations where a touch-screen is used for input.

How Links are Created

Suppose you were creating a database on the joys of pet fish. In a Hyperties™ article, you might write a sentence such as:

> Among the most interesting fish are ~guppies~ and ~goldfish~.

The special character ~ which surrounds the words *guppies* and *goldfish* is called a *tilde*. Placing tildes around a word or phrase, makes it a link. *The Author* program will automatically *resolve* each phrase surrounded by tildes by linking it to the article it references. *The Author* will ask for assistance when necessary. When *The Browser* displays your text, it removes the tildes and **highlights** the links:

> Among the most interesting fish are **guppies** and **goldfish**.

The form of highlighting depends upon the hardware the reader is using. On a monochrome monitor, the links are displayed in **boldface**, like the example above. On a color monitor the links are displayed in a different color from the rest of the text. You may select the colors if you wish by setting them in the TIES.SYS file. We provide a program called PALETTE to help you choose the colors.

Graphics and Videodisc

You may insert *graphics* in an article by adding the line:

@G=xxxxxxx

to the text (where xxxxxxx is the name you give to the graphic). For example, in the article about fish might include the sentence:

> The most beautiful fish are neon tetras which glow in radiant colors.
> @G=tetras

When Hyperties™ encounters the line "@G=tetras", it will locate the graphic stored on disk and display it.

You can prepare graphics for the Hyperties™ database using a standard graphics editor such as PC Paintbrush. Graphics may be any resolution (CGA, EGA, VGA). You may use a scanner to capture images and photos to be incorporated into the database. You can also create "text graphics" using the special characters available on the IBM PC; this is useful if charts and tables are to be displayed on computers which do not have graphics capability.

If you have an IBM Infowindow and the special Infowindow version of Hyperties™, you may insert videodisc sequences or still frames within articles. (*Contact Cognetics Corporation for more information*).

Graphic Backgrounds

You may create "graphic backgrounds" on which text is displayed. For example, you can use a "legal pad" or an "index card" as the background on which information is displayed. You may specify

a background to be used for the entire database or you may specify backgrounds for individual articles.

Graphic backgrounds can be used to mix text and graphics on a single screen. You can wrap text around an illustration.

To see graphic backgrounds, your reader needs a computer capable of displaying graphics. Although graphic backgrounds are possible in CGA 2-color mode, they are far less visually interesting and dynamic than EGA and VGA graphics. If your reader does not have an appropriate computer, Hyperties™ will simply display the text without the background.

We provide you with a set of backgrounds you can use, or you may create your own. See the Hyperties™ Tour for examples of graphic backgrounds.

Paths through the Database

You can provide your reader with four paths into the database:

- The **Index** which is automatically constructed for you, lists all the articles in the database and allows the reader to get to any specific one directly.

- The **Table of Contents** is a special article you may construct which to guide your reader to a specific topic or part of the database. Your reader can get to the Table of Contents at any time.

- The **Introduction** is a special article you can create which may be thought of as being at the "top" of the database. Your reader can begin browsing at the introduction.

- **Searching.** Hyperties™ allows the reader to search the contents of the database for articles containing a specified word or set of words.

The Introduction

When you create a Hyperties™ database, you can specify an Introductory Article. The Introductory Article usually contains the major links you wish to present to your reader and provides the reader with an overview of the database. There are a number of interesting techniques which can be used in the introductory article; see Chapter 6 -- *Authoring Strategies* for ideas. To create an Introductory Article, simply include an article named "INTRODUCTION" in your database.

The Index

Sometimes your readers will be looking for a specific article and will not want to browse the database starting with the introductory article. Hyperties™ automatically creates an index which lists all the articles in the database in alphabetical order. Your readers may go to the index at any time and access any article in the database directly. Because the index may contain a large number of entries, Hyperties™ allows the reader to turn to the index page for any specific letter. The index is accessed by selecting EXTRA from any article.

The Table of Contents

The alphabetical index may not always be the most efficient way for your reader to search the database of articles. Hyperties™ allows you to define a special article which serves as a *Table of Contents*. The table of contents may list the articles in any form you want; for example, by topics or grouped into chapters. To

create a Table of Contents, simply include an article named "CONTENTS" in the database.

Searching

Hyperties™ provides a string search capability which allows the reader to search the database for articles which have a particular word in them. The string search capability allows the reader to search for all articles which contain:

a specific word (example: *cat*)
a specific word followed by another word (example *tabby cat*)
any two words (example: both *cat* and *dog* in an article)
either of two words (example: *cat* or *dog* in an article)

The search process uses a special technique called *signature files* to perform the search, and is extremely fast.

When a reader requests a search, Hyperties™ goes through the database of articles and identifies each one which contains the specified words. It then constructs a list of all articles which contain the word or words. The reader may run through the list and view the brief description of each article on it. If the article looks interesting, the reader can view it by pressing ENTER, follow its links as desired, and then return to the list.

To activate the search capability, you run a special program we provide called UPDATE. This program creates the signature files for the database which make searching possible. The program need only be run once, but should be run again if any editing has been done to the database. The reader can search by selecting **EXTRA** from any article.

History

Hyperties™ maintains a history of the articles the reader has viewed. The reader may inspect this list at any time and may return to any article on the list directly.

The reader gets to the history from any article by selecting **EXTRA**.

Preface

Hyperties™ allows you to define a special article called "PREFACE", which you may use for instructions, author credits, or any other purpose you want. If your database contains an article called "PREFACE", *The Browser* will display it after the cover page.

Synonyms

You may want to refer to the same article using different words or phrases as links. This is often a matter of style. For example, suppose you were creating a Hyperties™ database about the presidents of the United States and you included an article on George Washington. Here are three sentences you might write (in the same or different articles) that link to the same article on George Washington:

> **George Washington** was the first president of the United States.
>
> A well-known anecdote about **Washington** involves a hatchet, a cherry tree, and his father's wrath.

> In understanding the political motivation of **the president**, it is important to consider his roots.

You can instruct Hyperties™ to treat **George Washington**, **Washington**, and **the president** as synonyms which all link to the same article. You do not need to plan this in advance; as Hyperties™ builds its index it will ask you if certain terms are to be considered synonyms or not.

Highlighting is Selective

Just because a word or phrase can be used as a link to an article, does not mean that it must be used that way every time it appears. As the author, you decide when a word or phrase is a link by enclosing it in tildes. This keeps articles from becoming cluttered with gratuitous highlighted terms. For example, you could write:

> ~George Washington~ was a military leader and as such commanded political respect. It was said that George Washington was autocratic and it was said that George Washington was a democrat.

When Hyperties™ displays this text, only the first reference would be highlighted as a link:

> **George Washington** was a military leader and as such commanded political respect. It was said that George Washington was autocratic and it was said that George Washington was a democrat.

This prevents your database from being cluttered up with distracting highlights.

Summary

Here is a summary of the main points covered in this chapter:

- Hyperties™ consists of two programs: *The Author* and *The Browser.*
- *The Author* is used to create a database of articles and graphics. These are linked together by words and phrases you designate.
- *The Browser* is used by your readers to access the Hyperties™ database. *The Browser* highlights the links in boldface or in color. When a reader selects a link, *The Browser* displays additional information -- a definition, a graphic, or a new article.
- Readers move around the database by means of the arrow keys, a mouse or (with a special version of Hyperties™) a touchscreen.
- *The Browser* keeps track of the reader's path and leads the reader back to the original article. *The Browser* will show the reader a history of all articles viewed.
- Hyperties™ constructs an index which the reader may use to access any article directly.
- You may define a special article as the Table of Contents.
- Your readers can search the contents of the entire database for articles which contain a particular word or two.

- You may define a special article as the Preface.
- You may designate one article as the Introduction. This is the first article the reader normally encounters.
- Hyperties™ articles may be prepared with the built-in word processor or on any standard word processing program. Graphics are prepared by standard graphics programs such as PC Paintbrush, or may be created by a scanner.
- You may create background graphics on which text is displayed.
- With a special version of Hyperties™ you may incorporate videodisc sequences into an article.
- You may declare different words or phrases to be synonyms which all link to the same article.

Chapter 2
The Browser

This chapter discusses *The Browser* which your readers use to access the Hyperties™ databases you create using *The Author*. *The Browser* works with any Hyperties™ database, so you may create many Hyperties™ databases and they will all work with the same Browser.

Since you will probably want to distribute the Hyperties™ databases you create, our license agreement permits you to make a number of copies of the Browser for each database. If you want to make more copies (for example, to distribute within a large corporation) or you want to re-sell the Browser with your database (for example, you are publishing an interactive Hyperbook), please contact Cognetics Corporation for an extended license.

Getting Around the Database

When the keyboard is used for browsing, only three keys are needed to access any article in the database (the left arrow key ←, the right arrow key →, and the ENTER key).

Your readers may also use a mouse to get around the database or (with a special version of Hyperties™) a touchscreen. For instructions on activating the mouse or touchscreen, see the section on the TIES.SYS file in Chapter 9.

As your readers become proficient in using *The Browser*, they may want to take advantage of a number of "hot keys" which can be used to get around faster than is possible with the arrow keys. Hot keys are discussed on page 37. However, we have designed *The Browser* so that readers do not need to know the "hot keys" to use all the program's features. The philosophy is that the novice or occasional user should not have to learn specialized commands, while the experienced user should have shortcuts available.

The Browser will give your readers the ability to:

- Begin reading the database starting with an introductory article you create.
- Read any article selected from the alphabetical index.
- Select an article from the Table of Contents if you choose to create one.
- Page forward and backward within an article.
- Obtain a brief description of any key words and phrases you chose to highlight within an article.
- View an illustration or videodisc sequence embedded in an article.
- While reading an article, link (jump) to another related article, by selecting a highlighted key word or phrase you specify. Hyperties[tm] will keep track of the path the reader takes and automatically lead them back to the point of departure.
- Review the list of all articles read during a session and revisit any desired. This is called *path history.*
- Search the database for articles containing a particular word or combination of two words (*search string*). Hyperties[tm] will construct a list of all articles containing the *search string* and the reader may view any or all of them.

Using The Browser

The best way to become familiar with *The Browser* is use it. To help you get started, and to show you some of *The Browser*'s capabilities, we have included a demonstration database called A Hyperties™ Tour with the programs.

You may run the Tour from a floppy disk or from your hard disk, if you have one. Performance will be more satisfying on a hard disk. If possible, run the Tour on a computer with color graphics capability (preferably EGA or VGA). On a color computer you will be able to see graphics and graphic backgrounds.

If you installed Hyperties™ using the **INSTALL** batch file supplied with the software, the Tour was automatically included on your hard disk.

To start the Tour from a hard disk:

1. Type the DOS command

   CD \HTIES

   to make the directory with Hyperties™ the current one.

2. Type the command

   BROWSER TOUR

   to start the Tour.

To start the Tour from a floppy disk:

1. Insert the floppy disk with the Hyperties[tm] Tour in the A: drive.

2. If necessary, type the DOS command **A:** to make A: the current drive. You will see the prompt **A>** on the screen.

3. At the **A>** prompt, type the command

   BROWSER TOUR

   to start the Tour.

Using a mouse:

1. To use a mouse to get around the tour, activate your "mouse driver" before starting *The Browser*. Then start *The Browser* with the command

   BROWSER TOUR /DM

   See chapter 9 for instructions on how to activate the mouse as a standard option.

The Cover Page

When you pick up a book, the first thing you see is its cover. Similarly, the first thing you see in Hyperties[tm] is a cover graphic. (You need a computer capable of displaying EGA graphics to see it).

When you create your own databases, you may create a cover for them. This cover may be text, any type of graphic, or you may omit it entirely.

Figure 2.1 below, shows the cover for the Tour:

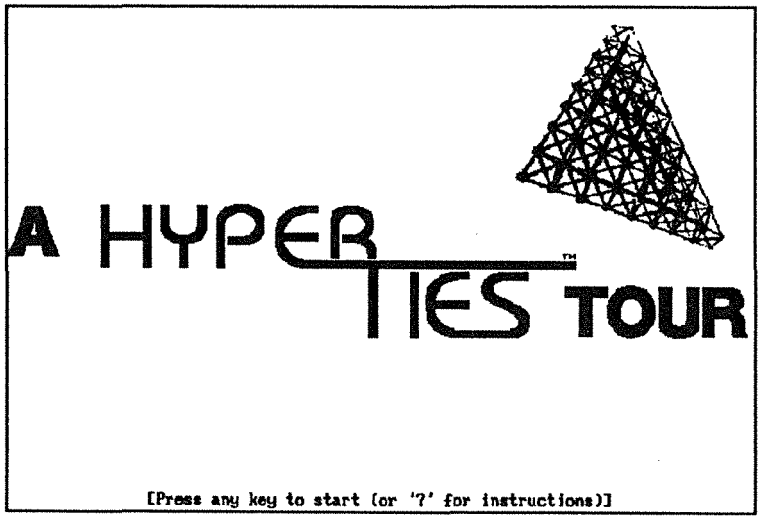


Figure 2.1
The Cover

At the bottom of the cover, is the message "Press any key to start (or ? for instructions)." If you press a ? you will see a help page. When you create your databases you may replace this help page with one of your own if you want.

The first page you will see (after the help page if you selected it) is the article called *Preface*. Figure 2.2 below, shows you the first page of the preface.

In your databases the preface is optional; you may use it for any purpose you want or omit it if you wish. In the Tour, we use the preface to create a little tutorial to help readers become familiar with Hyperties[tm].

If you are running the Tour on a computer with a color (EGA or VGA) monitor, you will see that the preface is displayed on a *graphic background* which looks like an index card. Graphic backgrounds can make your databases much more attractive and are easy to implement.

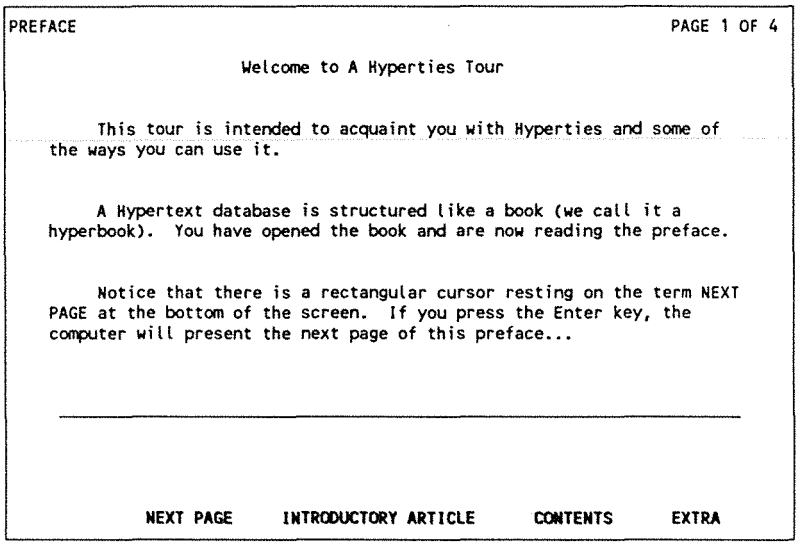


Figure 2.2
The Preface

Note the format of the article display. At the top is the *heading*. You set the heading in *The Author*. Also at the top is the current page number and number of pages (PAGE 1 OF 4).

The text appears in the center of the screen. We have used *embedded format commands* to format the text attractively.

At the bottom of the screen are the commands:

NEXT PAGE (go to the next page of the article)
INTRODUCTORY ARTICLE (go to the introduction article)
CONTENTS (go to the table of contents)
EXTRA (go to the index, path history and search)

The *selector bar* is resting on the command **NEXT PAGE**. You move the selector bar around with the arrow keys or the mouse (if activated). To execute a command, place the selector bar on it and press ENTER.

Use the selector bar to execute the **NEXT PAGE** command. This takes you to the next page of the article.

Highlighted Terms

When you are on the second page of the preface, you will see the additional command **BACK PAGE** on the bottom line. This allows you to page through the article in either direction.

The second page of the preface has a **highlighted term** in the text. The highlighted term is a *link* to additional information. It is the ability to link to additional information which is the basis of hypertext systems.

Place the selector bar on the highlighted term and press ENTER. You will see a *brief description* appear on the screen, between the text and the commands. This is shown in Figure 2.3 below:

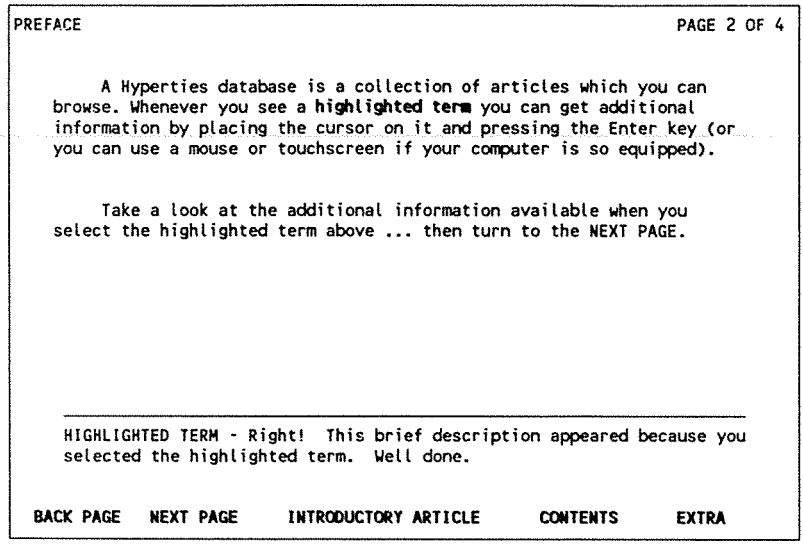


Figure 2.3
A Brief Description

When you create an article, using *The Author*, you may create a brief description. In some cases, a brief description is all you want to display. This is the case here. However, if you place the selector bar on **NEXT PAGE** and go to page 3, you will see a highlighted term which links to both a brief description and to another article.

Linking to An Article

The third page of the preface, has the highlighted term **complete article**. If you select it, you will get a brief description as before. But if you look at the bottom of the screen, you will notice a new command:

SEE ARTICLE: "COMPLETE ARTICLE"

This command informs the reader that a full article (text and/or graphics) is linked to the highlighted term and offers the reader the option of viewing it. Because the selector bar is positioned on the **SEE ARTICLE** command, you need only press ENTER to view it.

The reader is offered a choice! You need not go to the full article. The brief description should be written to help the reader decide.

For now, press ENTER to go to the article "complete article". You will see that it is one page long and therefore has no NEXT PAGE or BACK PAGE commands.

Note that the command **RETURN TO "PREFACE"** is on the bottom, as shown in figure 2.4, below. This is Hyperties[tm] way of keeping track of your path. To return to the article from which you came, simply place the selector bar on **RETURN TO** and press ENTER.

By the way, note that this article has no links in the text. You can only go back from this point. However, if this article had links, you could explore them, following an idea through many articles. This is the power of hypertext.

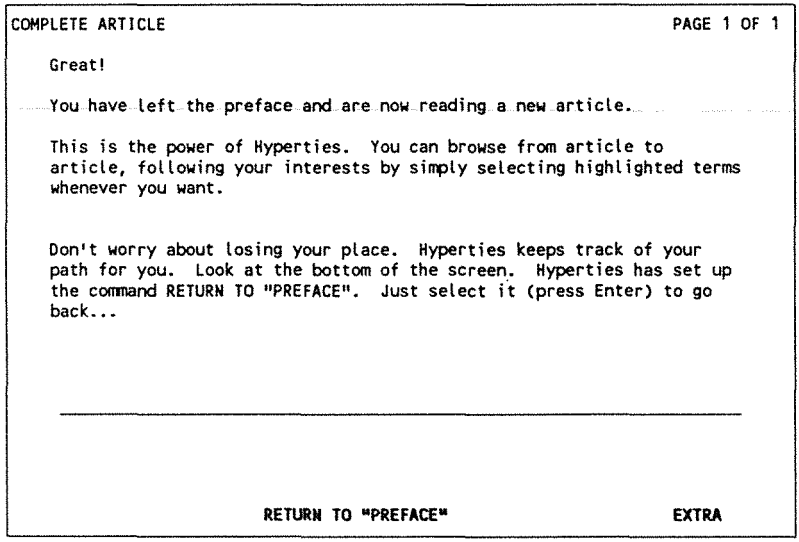


Figure 2.4
Linking to an Article

When you return to the preface, select **NEXT PAGE** to go to the fourth and final page of the article. Now you are ready to explore the commands at the bottom of the screen:

INTRODUCTORY ARTICLE, CONTENTS, and **EXTRA.**

Entering the Database

The TOUR provides you with three options for entering the database:

INTRODUCTORY - ARTICLE The introductory article is created by the author. It is sometimes called the *root* or *lead* article. The introductory article is designed to lead the reader to other key articles.

CONTENTS - The Table of Contents is created by the author and can be used to provide a topical index to key articles in the database.

EXTRA - The Extra command is used to get to the alphabetic index which lists all the articles in the database. From EXTRA, the reader can also get to the *path history* (a list of articles viewed), to *search* (locate all articles containing a specified word or two), and to QUIT.

The Introductory Article

Go to the introductory article and explore the database from it. Select links of interest to browse through the database; use the **RETURN TO** command to back up. (A shortcut -- you can also use the ESC key to back up one article).

Note that we created the introductory article as a single page. This facilitates a quick overview and provides the reader with a "home base" from which to explore the database.

There are many techniques for creating the introductory article; we discuss some in the chapter on *Authoring Strategies*. Although

most databases include an introductory article, you may leave it out if you wish.

The Table of Contents

When you have looked at some of the articles in the database, return to the preface and select **CONTENTS**. This will take you to the Table of Contents (You can use the RETURN TO command to back up until you return to the preface. You can also use the "hot key" ESC to back up one article -- this is the same as RETURN TO. The hot key CTRL-T will take you to the top of the database in one jump.).

The Table of Contents is an article which you may create in any form you want. In the Tour, we have used it as an index to important topics. Select a few articles starting with the Table of Contents to see how it works. Use **RETURN TO** or the ESC key to back up to it.

The Table of Contents is a convenient tool when your database has several topics, each beginning with a key article.

The Index

The index is an alphabetical listing of all the articles in the database. It is a convenient way to access any article directly. Every database has an index; it is constructed automatically by Hyperties™.

To get to the index, select **EXTRA** from any article.

Figure 2.5 below shows a page from the index. Note that it is designed to look like a file folder with tabs at the top. Like an article, there are commands at the bottom of the screen:

NEXT PAGE (except on the last page), **BACK PAGE** (except on the first page), **RETURN TO** and **QUIT**.

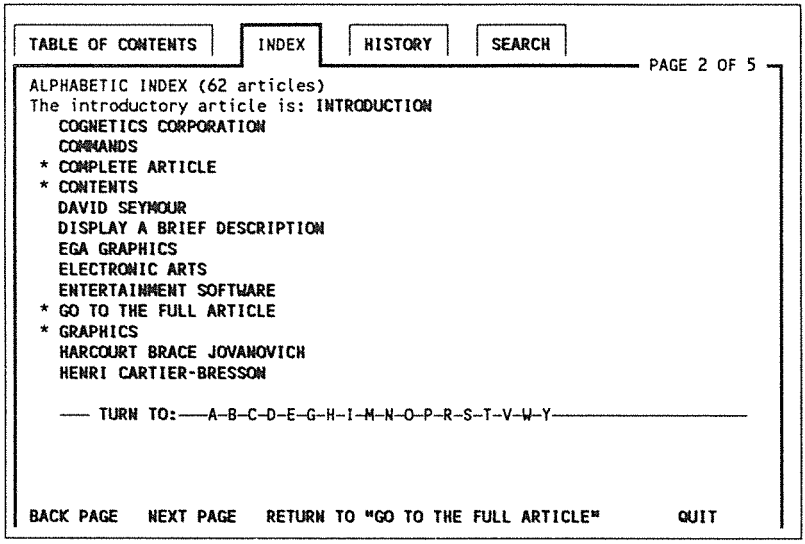


Figure 2.5
The Index

Note that the selector bar can be moved up and down. To select any article, place the bar on it and press ENTER. You will get the brief description. If there is a full article, you will get the **SEE ARTICLE** option, just as if you had selected a highlighted term in the text of an article.

Use **NEXT PAGE** and **BACK PAGE** to page through the index. If you want to page more quickly, place the selector bar on **TURN TO** and press ENTER. Place the selector bar on the letter you want to turn to and press ENTER again. You will go directly to the appropriate page of the index and the selector bar will land on the first article beginning with the letter you specified.

From the index you can also get directly to the Table of Contents and to the Introductory Article. You can also access two additional features: **HISTORY** and **SEARCH**. These are discussed in the sections which follow.

History

You get to *history* by selecting it from the index. Place the selector bar on the tab labeled **HISTORY** and press ENTER.

You will see a display like the one shown in figure 2.6 below:

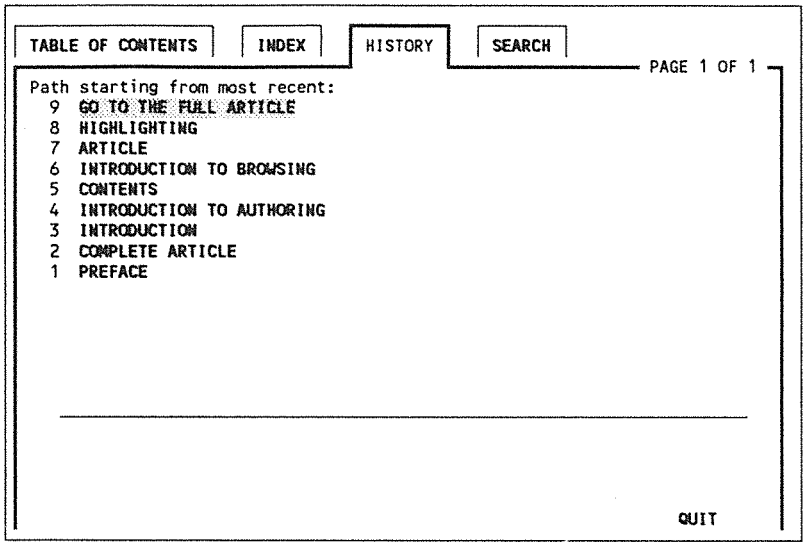


Figure 2.6
The Path History

The display shows a list of the articles you have visited. Since the list depends upon your activity, it will differ from the one shown in figure 2.6.

The list shows the most recent article visited at the top. Visits to the index are not recorded in the list but visits to the Table of Contents are.

You may re-visit any article on the list by placing the selector bar on it and pressing ENTER.

Searching

Searching is one of the most powerful ways of using a Hyperties™ database. To search, select **SEARCH** from the index. You will see a screen with the selector bar on **SEARCH STRING**. Press ENTER, type in the string "museum", and press ENTER again.

The computer will search through every article in the database looking for the word "museum". It will then display a list of all articles which contain the word. Figure 2.7 below shows an example of a search:

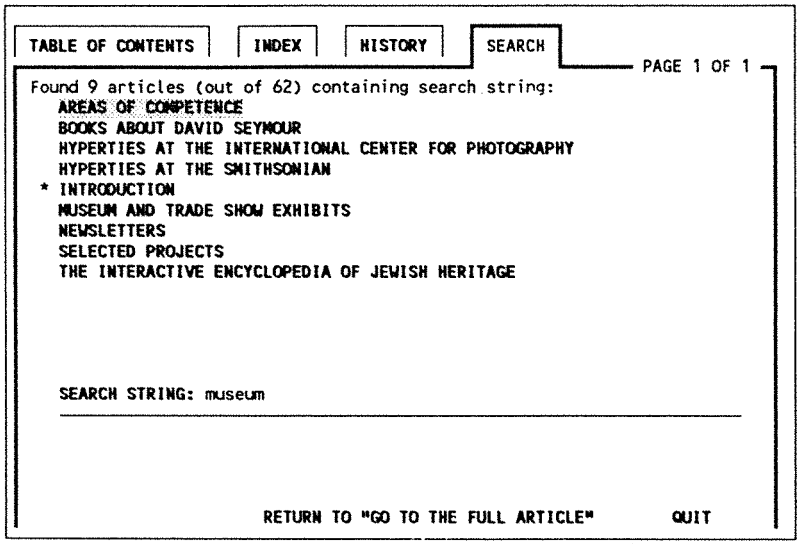


Figure 2.7
Searching

As you will see, the search is remarkably rapid, although it depends upon the speed of your computer and disk to some extent. On a fast machine, it is almost instantaneous.

Once you have the list, you can select any article on it and read the brief description. If the article looks relevant, you can visit it. When you visit an article from the search screen, you will go directly to the page which contains the word you searched for. The word will be highlighted so you can see it easily.

Searching for Two Words

You can search for a single word as described above. You may also enter one of the following as your search string:

word1 word2	This searches for articles which contain *word1* followed directly by *word2*.
word1&word2	This searches for articles which contain both *word1* and *word2*.
word1|word2	This searches for articles which contain either *word1* or *word2* or both.

The search function will ignore common words such as "and" or "the", since these appear with too much frequency. If you attempt to search for one of these common words, the system will indicate that it is too common to serve as a search term.

Note that you *cannot* search on phrases like "the first president". This would be valuable but would slow the search process. The

technique used is very rapid but somewhat limited. This is typical of computer solutions.

Printing

You can print the article you are currently reading by holding down the *Ctrl* key and pressing *p* (that is, *Ctrl-p*). If you are at the index, *Ctrl-p* will print the entire index.

Hyperties™ assumes that you have a printer attached to LPT1: on your computer.

To cancel a print request, press *Ctrl-c*.

Moving On

By this point you should have become a fairly sophisticated user. Experiment with *The Browser*. Look at various articles and explore the index and search capabilities. By experimenting with The Browser you will become adept at using Hyperties™ in a very short time.

As you go through the Tour, you will see more features of Hyperties™. Look at the various graphics and graphic backgrounds. See how CGA and EGA differ. The Tour does not contain any VGA graphics but Hyperties™ does support them.

Look at the article titled "imbedded illustrations". This article uses text overlaid on graphic backgrounds to create graphics with highlighted terms or captions. This is a very powerful technique.

Hot Keys

The Browser provides a number of "hot keys" which can speed up the process of getting around a database. You never need to use these keys but they are available if you want them.

You can get to a highlighted term directly by pressing its first letter. If several terms on the screen begin with the same letter, pressing it will cycle around them.

The following hot keys can also be used (when *Ctrl* is specified it means, hold down the Ctrl key and press the letter key):

Esc	return to the previous article
Pg Up	back page
Pg Dn	next page
Home	go to the first page of an article
End	go to the last page of an article
Ctrl-t	go to the first article you saw (the TOP)
Ctrl-e	go directly to the index (same as EXTRA)
Ctrl-p	print the current article or index, depending on where you are
? or *Ctrl-h*	display the help page
Ctrl-q or *F10*	quit *The Browser*, return to DOS (or to *The Author* if you were authoring).

Summary

Here is a summary of the main points covered in this chapter:

- *The Browser* is the program which enables your readers to use the Hyperties™ databases you create.
- To use *The Browser*, your readers need only position the selector bar on a highlighted term and press ENTER.
- Highlighted terms which appear at the bottom of the screen are commands.
- Highlighted terms which appear in the text of the article are links to additional information.
- A Hyperties™ database can have a cover.
- A Hyperties™ database can begin with *preface* if you wish.
- When you begin reading a Hyperties™ database, you may start at the *introductory article, the Table of Contents*, or you may go to the index and select any article directly.
- When you select a reference, Hyperties™ shows you a brief description. If there is article content linked to the reference, Hyperties™ will offer the option of leaving the current article and visiting the referenced article.

- When visiting a new article, Hyperties[tm] places a "bookmark" in the current article to hold your place, so you can return.

- Hyperties[tm] provides you with a path *history* showing you a list of all articles visited.

- Hyperties[tm] provides a *search* capability which can search the content of the entire database for articles containing a particular word or set of two words.

- Background graphics can be used to display the text on a background such as an index card or pad. Background graphics can also be used to overlay text with highlighted terms on a graphic.

Chapter 3
The Author

This chapter describes *The Author* (version 2.3) which is the program you use to create and modify a Hyperties™ database - Chapter 6 is a tutorial which walks you through the authoring process. *The Author* includes a built-in word processor which allows you to create articles. If you prefer, you may use your own word processor and "import" articles. You create graphics separately using a program like PC PAINTBRUSH and reference them within articles. If you have the videodisc version of Hyperties™, you may also reference videodisc frames.

Using the Author Tool

The easiest way to learn to use *The Author* is to work with it. The organization of *The Author* is simple and consistent and it always shows you what options are available to you. The important thing is to comprehend its basic structure, once you have done this you will find it an easy-to-use tool.

The Author will ask you to confirm any command which might do something drastic to your database (such as deleting an article). Therefore you should feel free to experiment with compiling your Hyperties™ database.

An Overview of Authoring

1. Use the MKDIR command in DOS to create a directory in which you will store your articles. You may store graphics in the same directory as the articles or you may create a separate one. If you plan to store graphics in a separate directory, create one for them.

2. Issue the CD command to change to the directory in which *The Author* is located, for example:

CD \HTIES

Then start *The Author* specifying the directory in which you will store your database:

AUTHOR *directory*

The Author will begin on its index screen. There will, of course, be no articles in the index yet.

3. Use the **ADD** command to place the first article in the index. Then use the **EDIT** command to go to the *edit screen* where you will create or import it.

4. Create the heading, brief description, and content (if you want) of the article. Alternatively, use the **GET** command to import them.

   If there are references (links) to other articles in the content, enclose them in tildes (e.g. This sentence is ~about Hyperties~.).

   If you want to insert illustrations in the article use an **@G=** command. For videodisc sequences use the **@V=** command. For graphic backgrounds use the **@B=** command. If you want to control the format of the article, use embedded format commands as described in Chapter 4.

5. Use the **RESOLVE** command to save the article and insert the links (if any) into the index.
   The Author will ask you for help in resolving. When resolution is complete, you will be returned to the index screen. From the index screen you can **BROWSE** to look at the article.

6. Repeat the process of creating and resolving articles until your database is complete.

7. If you want a preface (instructions or author credits), create an article called "PREFACE".

   If you want a table of contents, automatically accessible from the INDEX and the Preface (if any), create an article named "CONTENTS".

   If you want an introductory article, also automatically accessible from the INDEX and Preface, create an article named "INTRODUCTION".

8. To make SEARCH possible, run the program UPDATE to create the signature files used by *The Browser.*

9. Create the graphics you want for your database including a cover (if you want). Create a TIES.SYS file if you want to change the standard colors or other elements used by *The Browser* (see Chapter 9 on the TIES.SYS file for details).

Starting the Author Tool

If you are creating a new database, create a directory for it. If you are modifying an existing database, you can just start the program.

Assuming you installed Hyperties[tm] on a hard disk, as specified in the chapter on installation:

1. Use the MS-DOS command CD (change directory) to change the directory to the one in which you have stored the Author Tool. For example:

   CD \HTIES

2. Type the command:

   AUTHOR *xxxxxxx*

   and press the ENTER key. Substitute for *xxxxxxx* the directory name of the Hyperties[tm] database you are creating or modifying. For example, to create or modify a database on the hard drive c: in the directory \POLICY, you would type:

   AUTHOR C:\POLICY

To start *The Author* from a floppy disk:

1. Insert the disk with *The Author* in floppy drive "A:"

2. If necessary type the MS-DOS command "A:" so you get the prompt **A>** on the screen.

3. Type the command:

   AUTHOR *xxxxxxx*

and press the ENTER key. Substitute for *xxxxxxx* the directory name of the Hyperties™ database you are creating or modifying.

WARNINGS:

1. Be careful how you use backslashes. The command:

   AUTHOR \MYSTUFF

   specifies that your Hyperties™ database is in the directory **C:\MYSTUFF** while the command:

   AUTHOR MYSTUFF

   specifies that your database is in the directory **C:\HTIES\MYSTUFF**. If you are confused by this, we suggest that you read about directories in the manual which came with your copy of DOS or consult someone experienced with directories.

2. If you are creating a database on a floppy disk, we do not recommend putting it in the root directory. The reason is that DOS limits the number of files which can be stored in a root directory and you may bump up against that limit. Therefore, do not create the database in (for example) **B:** but in **B:\MYDATA**.

3. If you are creating a new database, use the MD (Make Directory) command to create a directory for it, before starting *The Author.*

4. If you are creating graphics and storing in a separate directory from the articles, create a TIES.SYS file with the line:

   graphics = xxxxxxxx

   substituting the directory containing the graphics for xxxxxxxx. If you don't do this, neither *The Author* nor *The Browser* will be able to find the graphics. See Chapter 9 for details.

Articles

The basic units in a Hyperties[tm] database are *articles*, linked together. Therefore, in *The Author* you will be creating and editing articles. An article consists of the following elements:

- *The Index Entry* (required). This is the name by which the article is entered in the alphabetical index. Articles are entered into the index in two ways: explicitly when you **ADD** them from the index screen, and automatically during the **RESOLVE** process.

- *The Heading* (optional but usually presented except when you need the space for part of a graphic background). When the article is displayed by *The Browser*, the heading appears on the top line in the left-hand corner. *The Author* initially assumes that the heading is to be the same as the index entry. If you wish, you may change it.

- *The Brief Description* (optional but often used). The brief description can be as long as 150 characters (but must fit on two lines). When the reader selects

a link, the brief description will be displayed. The brief description may be used to define a term or to help your reader decide whether to go to a full article on the referenced topic. If the brief description is sufficient, you need not write a full article.

Hyperties™ provides several options for handling the brief description which are discussed below.

- *The Content* (optional). The content of the article may include text (with *links* to other articles), formatting commands (which let you control the way text is displayed by *The Browser*), references to graphics, references to videodisc sequences, specification of a graphic background on which the text is to be displayed, specification of other articles to be included within the current one.

  If you create article content, Hyperties™ will offer your reader the option of viewing it when the corresponding reference is selected.

- *Notes* (optional). *The Author* lets you keep notes on each article. These notes are for your own use and are not shown to the reader.

Executing Commands

You get around within *The Author* by using the *selector bar* exactly as in *The Browser*. Commands are listed on the bottom of the screen. To execute a command, use the arrow keys to place the *selector bar* on it and press ENTER. As with *The Browser*, you may press the first letter of the command (instead of the arrow keys) to jump directly to it. If several terms begin

with the same letter, press the letter key until the selector bar is on the action you want, then press ENTER.

If you are inside a window (for example, the list of articles in the index) press ESC to leave it.

When you become experienced in using *The Author* you can execute any command directly by holding down the **Alt** key and pressing the first letter of the command. For example **ALT-E** to EDIT an article. We suggest you become familiar with *The Author* before using this shortcut.

The Index Screen

The index screen, shown in figure 3.1 below, is your "home base" when authoring. The index screen consists of a window in which all the articles in the database are listed. Commands are listed at the bottom of the screen.

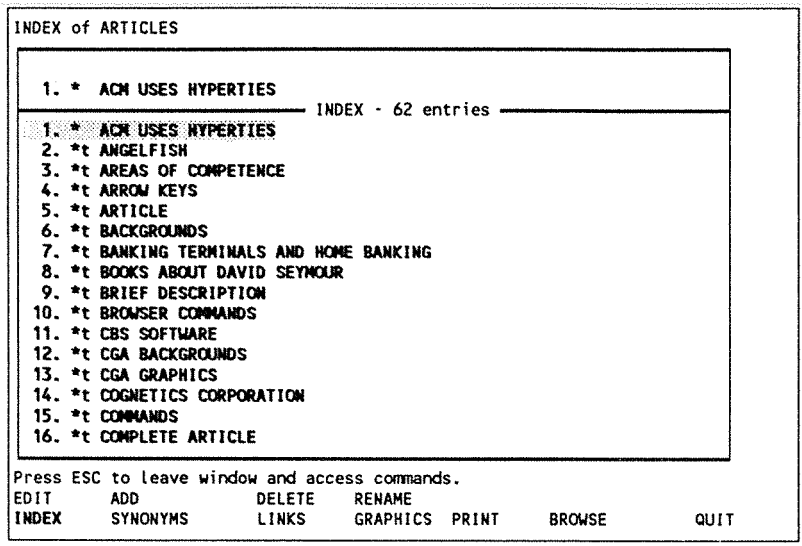


Figure 3.1
The Index Screen

When you begin creating a new database, the list of articles will be empty. Once the database contains articles, they will be listed in alphabetical order.

The article on which you are working (the "current article") is displayed in the small window at the top of the list. As you move the selector bar up and down the list of articles, the current article will change. You move the selector bar with the arrow

keys or you can use one of the following keys to move around the list of articles:

scroll the list up a full page	*Pg Up*
scroll the list down a full page	*Pg Dn*
move to the beginning of the list	*Home*
move to the end of the list	*End*
leave the window	*Esc*

If the article list is long, moving the selector bar up and down could be tedious. Therefore you can jump directly to any article by typing its name. When you are in the window listing the articles, simply start typing and you will automatically go into "search mode" in the *current article* window. You need not type the whole name. Just type a letter or two. When you press ENTER, the selector bar will jump to the article which is closest to what you typed.

To leave the window and execute another command, press ESC (or press **ALT** plus the first letter of the command). To re-enter the window, execute the **INDEX** command (or press **ALT-I**).

These commands work with any window containing a list of items.

Commands on the Index Screen

You may execute any of the following commands from the index screen by placing the selector bar on it and pressing ENTER (press ESC first if you are in the window) You can also press **ALT** plus the first letter of the command:

ADD

Add a new article to the index. When you add an article, *The Author* will ask you to type its name. When you press ENTER, the article will be added to the index, in its proper place. Press ESC to cancel the ADD.

DELETE

Delete the current article. *The Author* will ask you to confirm the deletion before it actually does it.

RENAME

Change the name of the current article. You will be asked to type in the new name, and then confirm it.

EDIT

The EDIT command takes you to the edit screen where you can create or edit the article. See the section *Creating and Editing Articles* below for details.

LINKS

The LINKS command takes you to a screen which shows you all the links from- and to- the current article. If the number of links is more than will fit in the display window, you can scroll it using the keys listed for the index.

SYNONYMS

The SYNONYMS command takes you to a screen from which you can view all the synonyms for the current article.

INDEX

The INDEX command is used to place you back inside the window when you have left it by pressing ESC.

BROWSE

The BROWSE command calls *The Browser* at the current article. This allows you to see how the article will look to your readers. For this command to work, *The Browser* must be in the same directory as *The Author* or it must be accessible through the DOS PATH command. When you QUIT the Browser (by selecting QUIT from the index or pressing CTRL-Q) you will be returned to *The Author.*

PRINT

The PRINT command prints the index, listing all the articles in it and their synonyms.

GRAPHICS

The GRAPHICS command shows you a list of all the graphics currently in the database. These are either the graphics in the directory with the articles (that is, the one you specified when you started *The Author*), or those in the graphics directory specified in the TIES.SYS file (if you specified one).

To display any graphic, simply place the selector bar on it and press ENTER. Press any key to return to the original display.

QUIT

The QUIT command is used to exit *The Author.*

Creating and Editing Articles

To create or edit and article make it the current article on the index screen by placing the selector bar on it. (If you are creating a new article, **ADD** it to the index). Then execute the **EDIT**

command. You will go to the *edit screen* shown in figure 3.2 below.

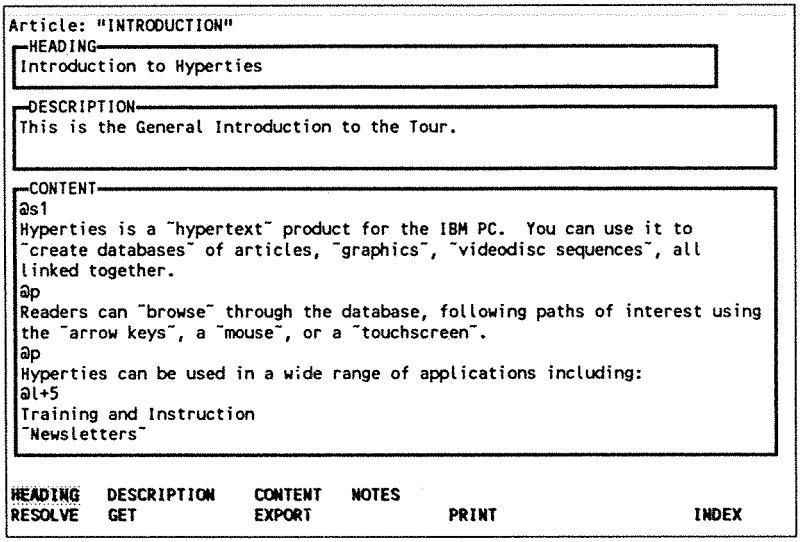


Figure 4.2
The Edit Screen

As you can see from figure 3.2, the *edit screen* is similar to the *index screen* in structure. It contains three windows labeled **HEADING, BRIEF DESCRIPTION**, and **CONTENT.**

Use the **HEADING** window to enter the heading or title of the article. The heading is displayed by the Browser when your reader selects the article. Usually the heading of the article is the same as its entry in the index. Therefore when you first come to the *edit screen*, Hyperties™ will fill-in the **HEADING**

window with the index entry. However you may change it if you wish.

You might want to change the heading, for example, if you were writing an article on George Washington. In the index, you might want the article to be listed as WASHINGTON, GEORGE so a reader will find it under the letter *W*. However, it might look better to display the article with the title GEORGE WASHINGTON. The choice is yours.

You may also want to eliminate the heading to use the space for a graphic background. For an example of this, see the "Cognetics Corporation" article in the Hyperties™ Tour.

To enter the HEADING window execute the **HEADING** command at the bottom of the screen (or press **ALT-H**). Press ESC to leave it.

The second window is labeled **BRIEF DESCRIPTION** and is where you enter the brief description of the article.

To enter the BRIEF DESCRIPTION window execute the **DESCRIPTION** command at the bottom of the screen (or press **ALT-D**). Although the text in this window will scroll beyond two lines, DON'T enter more than two lines of text in your brief description. Press ESC to leave it.

The third window is labeled **CONTENT** and is the place where you enter or edit the text of the article. To enter the CONTENT window execute the **CONTENT** command at the bottom of the screen (or press **ALT-C**). The section *Entering and Editing Text* below discusses the built-in editor.

Other commands which appear on the *edit screen* are:

NOTES

The NOTES command opens a window on which you can record notes for your own use. Notes can only be read by the author; they cannot be accessed from *The Browser.*

RESOLVE

The RESOLVE command is used to save the article and resolve the links to other articles. It is the normal way of leaving the *edit screen.* See the section on *Resolving Links,* below for more information. To exit the *edit screen* without resolving, use the INDEX command.

GET

The GET command is used to import an article created on another word processor. See the section on *Importing and Exporting Articles* for details.

EXPORT

The EXPORT command is used to export an article as a "clean ASCII" file so it can be read by other word processors or imported into another Hyperties™ database. See the section on *Importing and Exporting Articles* for details.

Entering and Editing Text

If you are entering or editing text using the built-in word processor, you may use any of the following commands:

ACTION	KEYS
move the cursor around the text	use the four arrow keys
delete previous characters	use the backspace key (←) located above the ENTER key
delete the current character	*Del*
toggle between insert & overwrite	*Ins*
scroll up a "page" of text	*Pg Up*
scroll down a "page" of text	*Pg Dn*
move to the beginning of the line	*Home*
move to the end of the line	*End*
insert a line	*ENTER*
delete a line	*Ctrl + D*
move to the beginning of the text	*Ctrl + Home*
move to the end of the text	*Ctrl + End*

(continued on next page...)

ACTION	KEYS
move to the next word	*Ctrl* + → *(Right Arrow)*
move to the previous word	*Ctrl* + ← *(Left Arrow)*
leave the text window (done editing)	*Esc*

When you enter text using the built-in word processor you will get automatic word wrap. If a word does not fit on the current line, it is automatically moved to the beginning of the next line. When you enter a carriage return (by pressing ENTER) you will see it as a small arrow (←). *The Browser* will not display carriage returns.

Specifying Links

You specify references to other articles by enclosing the word or phrase to be highlighted within tildes (~). When the tilde key is pressed, a *pair* of tildes will appear on the screen. As you type the word or phrase to be highlighted, the tildes will expand to surround it (as long as you are in *insert* mode). After the entire term to be highlighted is typed, move the cursor past the closing tilde and continue entering the text.

If you are using your own word processor and importing the text, simply place tildes around the word or phrase you want highlighted.

If you use your own word processor, you can use automatic "search and replace" to place tildes around words or phrases you want highlighted.

Formatting the Display of Text

You may embed formatting commands in the text of an article to control such elements as *line spacing*, *paragraph indentation* and *margin width*. All embedded commands use an @ followed by a letter (for example @N to force a new page). See Chapter 4 for a complete list of formatting commands.

Embedded commands are also used for inserting graphics (@G), for specifying backgrounds (@B), and for specifying videodisc frames (@V). These are discussed in Chapter 5.

Importing and Exporting Articles

If you already have text files on disk (for example, memos or letters) which you want to integrate into your Hyperties[tm] database, you may "import" them. The files you import must be in "clean ASCII" format. Clean ASCII means that the files consist of standard text characters. Many word processors, such as Word Perfect and MultiMate embed hidden control codes in the document to perform such functions as centering, tabs, and end of page. Such files are *not* "clean."

Fortunately, these word processors have an option to create a clean file. For example, in Word Perfect you use CTRL-F5 (Text In/Out) to save your file as a DOS text file. In MultiMate there is a convert file utility which will save your text as an ASCII file. Consult your word processor's manual on how to do this.

There are also powerful editing programs such as BRIEF which are oriented toward programming and which normally produce clean ASCII files.

To import a file into Hyperties™:

1. Go to the *edit screen.*

2. Select the **GET** command on the edit screen.

3. You will see the message:

   "Type in the name of the file to import. Then press ENTER."

4. Type in the file name you want to import. Use the "full path name" if required. Example:

   A:\WORDS\ARTICLE.TXT

   Press the ENTER key to import the article. Press the ESC key to cancel the import.

When you import an article you may want to import the heading and brief description as well. To do this, you may embed the following commands in the text file: **@HEADING**, **@DESCRIPTION**, and **@CONTENT**. These commands must appear in the correct order: first heading (if used), then description (if used), and finally, content. If you don't use these commands, the text is assumed to be content only.

> **Important note:** Make sure that any imported *brief description* is less than 150 characters. A longer brief description is possible, but runs the risk of giving *The Browser* serious problems.

Here is an example of how the embedded commands can be used in a file to be imported:

> @HEADING
> George Washington
> @DESCRIPTION
> First President of the United States
> @CONTENT
> George Washington was the first President of the United States. He was married to ~Martha Washington~. He made his home in Virginia; his estate was called ~Mount Vernon~. There are several portraits of George Washington painted by ~Gilbert Stuart~, one of which is reproduced on the One Dollar bill.

The Author will also export articles, creating "clean ASCII" files which can be read by word processors. When an article is exported, the heading, brief description and content will all be exported, prefixed by **@HEADING**, **@DESCRIPTION**, and **@CONTENT**. This permits you to export a file, edit it on a word processor, and then import it again.

To export a file, select **EXPORT** from the edit screen. You will see the message:

> Type in the full name of the file to export, then press ENTER.

Type in the name of the file to which the article should be saved. Use a full pathname if needed. Example:

> **A:\DOCS\ARTICLE.TXT**

The article will be saved. If the file already exists, you will be asked if you want to overwrite it.

Summary

Here is a summary of the main points covered in this chapter:

- The Author is the program which you use to create a Hyperties™ database (hyperbook).
- The basic units of the Hyperties™ database are *articles* which consist of a heading, a definition, and (optionally) content, which consists of text, references to illustrations and videodisc sequences, and formatting commands.
- Every article is entered into the *index*, either explicitly by the ADD command or as a result of resolving references to it.
- The two key authoring screens are the *index screen* and the *edit screen*.
- Text may be entered using the built in word-processor or by importing text produced on an external editor.
- References (links) to other articles are specified by enclosing them in tildes (~~).
- Embedded format commands are used to specify the presentation format to the Browser.
- Illustrations (graphics) are referenced by the @G= embedded command. Videodisc sequences by the @V= command. Backgrounds are specified by the @B= command. These commands are discussed in the chapter on graphics.

Chapter 4
Format Commands

You may embed formatting commands in the text of an article to control such elements as *line spacing*, *paragraph indentation* and *margin width*. If you are entering tabular information or overlaying text on a graphic background, you can use *table mode* which allows you to specify precise display formats.

Embedded Formatting Commands

Embedded formatting commands begin with the character @, followed by a letter (which can be lower or upper case). For example **@p** or **@P** is an embedded command.

Some commands require additional information following the letter. For example, to set line spacing to 2 (double space) the command **@S2** (or **@s2**) is used. To move the left margin 5 spaces inward, you could write **@L+5**. The details on each embedded format command are given below. For the Browser to recognize that a line contains embedded formatting commands, the @ must be in column 1. We recommend that each command appear on a separate line to improve readability; however, you can place more than one on a line.

The following example sets single spacing and a left margin of 10 for the first two lines of text; double spacing and a left margin of 15 for the next two:

```
@s1
@L10
Line 1 of text
Line 2 of text
@L15
@s2
Line 3 of text
Line 4 of text
```

Standard Settings

In the absence of embedded formatting commands, *The Browser* will double space all text, display text using a left margin setting of 5 and a right margin of 75. This produces a nicely formatted display in which the text of the article is indented below the heading line.

The Browser will automatically "word wrap" text to fit the margins. Wrapping is performed on whole words only; words are never split between lines.

Highlighted items are never split because the highlight bar would look odd if it were divided between two lines. You may have to do some fine tuning of the phrasing of your text to make long highlighted terms look good. The BROWSE option on the *index screen* can be used to see how an article will look when displayed by *The Browser.*

While word wrap is very convenient in most cases, it can be undesirable when you have text you want displayed exactly as you entered it. This usually is the case when you are creating tables or overlaying text on a background graphic. For these situations, Hyperties[tm] provides a special *table mode* command which instructs the Browser to display the text exactly as entered.

If the embedded format command is in error (for example, margins that exceed the screen window), *The Browser* will ignore them and use the standard settings.

Single and Double Spacing

@S - Line Spacing

This command is always written as **@S1** (**@s1**) for single spacing or **@S2** (**@s2**) for double spacing. All subsequent lines in the article will continue to be displayed with the requested spacing until another @S command is encountered.

Force a New Page

@N - New Page

This command instructs *The Browser* to start a new page.

Margin Setting Commands

Margin setting commands require that margin calculations not exceed the physical screen limits of 1 to 80 columns. A minimum of 20 spaces between the left and right margins is required.

@L - Set Left Margin

This command is used to set (or re-set) the left margin. The standard setting places the left margin at column 5, which provides a nice looking indent relative to the heading line.

With the left margin command you may specify the starting column number. For example, to begin the left margin in

position 10, you would write **@L10**. To begin it in position 15, you would write **@L15**. When the command is used in this way, each new **@L** command resets the left margin. For example:

```
@L10
This text will begin in column 10.
@L15
This text will begin in column 15.
```

The second way to use the command is to specify the number of positions the margin should be shifted, relative to its current position. For example, to indent the left margin five spaces in, you would write **@L+5**. To move the left margin five spaces out, you would write **@L-5**. When the command is used in this way, each time the Browser encounters an "@L", it resets the left margin adding or subtracting the number from the current margin position. For example:

```
@L10
This text will begin in column 10.
@L+15
This text will begin in column 25.
```

Be careful, when specifying a negative number, not to attempt to shift the left margin past column 1; for example if the current margin setting were at position 10, it would be *incorrect* to write @L-15; in this case *The Browser* would return to the standard setting.

@R - Set Right Margin

This command is analogous to the command used to set the left margin but affects the right margin. The standard setting is at column 75; you can shift the right margin inward (*-n*) or outward (*+n*).

For example, a command of **@R-5** will shift the right margin 5 spaces inward (to the center). A command of **@R+4** will shift the margin outward four spaces. Be careful not to shift the margins past the edge of the screen (column 80), this means that from the standard margins, you should not shift the right margin more than +4. Also remember that there must be at least 20 spaces between the left and right margin.

@M - Set Both Margins

The **@M** command is used to shift both the left and right margins simultaneously. This is often used to indent quotations.

Specify the number of spaces to shift, relative to the current margins. For example the command **@M+5** can be used to shift both margins inward 5 spaces. The command **@M-6** will shift both margins outward 6 spaces. The **@M** command is particularly useful when you want to indent a quotation:

> For was it not the poet who said:
> @M+8
> Twinkle, twinkle little star. How I wonder what you are.
> @M-8
> And have we not all felt the same thing?

Table Mode

@T - Table Mode

The **@T** command is used to turn *table mode* on and off. When table mode is *off* all tildes (~) in the text are deleted. However, when table mode is *on*, all tildes (~) are replaced by blank spaces rather than merely deleted. This enables you to line up columns easily.

For example *The Browser* would display the following command and text sequence:

```
@T
     Department    Supervisor     Product

     Marketing     ~Newton~       words
     Research      ~Einstein~     numbers
@T
```

as follows:

Department	Supervisor	Product
Marketing	**Newton**	words
Research	**Einstein**	numbers

In table mode, all formatting commands previously specified are disregarded and you have total control over the display format. In table mode, lines are always single spaced. In order to double space lines enter blank lines in the text.

Starting a New Paragraph

The embedded formatting command @P instructs *The Browser* to begin a new paragraph. You can also begin a new paragraph by inserting two carriage returns in the text and indenting the first line of the next paragraph if desired. Assuming that ← represents a carriage return, you can format a paragraph as follows:

This ends the previous paragraph←
←
    This line begins a new one.

The command **@P** forces the next paragraph to begin on a new line and optionally indents the first line of the next paragraph. The example above could also be written:

This ends the previous paragraph.
@p5
This line begins a new one.

The command @P without a number following, ends the current paragraph but does not indent the next line. Successive **@P** commands add blank lines to the text.

If you wish to indent paragraphs, you can also write this command as **@P***n* where you substitute a number between 1 and the width of the line for *n*. The next line will be indented *n* spaces.

Including Articles

@A - Include another article in this one.

The **@A** command is used to include the text of another article within the current one. The heading and brief description of the article specified are not included.

The format of this command is

@A=xxxxxxxxx

or

@A="xxxxxx xxxxx"

the quotes are required if the article name has a space in it. Substitute the name of the article to be included for *xxxxxxx*.

Here is an example:

Many tropical fish are good pets.
@A=TETRAS
@A="ANGEL FISH"
Some are not, however:
@A="SIAMESE FIGHTING FISH"

Setting Colors

Within *The Author*, you can set the colors used for text, for highlighted links, for the selector bar, and for links or commands which have been selected (the link changes color to let you know that the program is responding to your command). To set colors for an entire database, see Chapter 9. To set colors within an article, use the following embedded commands:

@CT=*nn* - Text color: a decimal number from 0 to 15.

@CH=*nn* - Highlight color: a number from 0 to 15.

@CC=*nnn* - Cursor color (selector bar): a number from 0 to 127 (background color [0 to 7] times 16, plus foreground color of 0 to 15).

@CS=*nn* - Selection color: a number from 0 to 15.

In all cases, the numbers are the same numbers specified by IBM and used in BASIC and DOS. For help is determining the right

numbers for custom text colors, both in general and for individual articles, we have supplied the **PALETTE** program. You can find the program on the *Author & Utilities* diskette; instructions for its use are in the *Addendum* at the end of this manual.

Other Embedded Commands

Other embedded commands are discussed in the chapter on graphics. They are:

@G - Include a graphic illustration.

@V - Include a videodisc sequence.

@B - Display the text on a graphic background.

Summary

Here is a summary of the main points covered in this chapter:

- Embedded formatting commands can be used to control the display of text.
- @P is used to begin a new paragraph.
- @L, @R, @M are used to set the margins.
- @N is used to force a new page.
- @T places you in table mode.
- @A is used to include the text of an article within another one.

Chapter 5
Graphics and Videodisc

This chapter describes how to insert graphics into an article. Graphics may be illustrations which are a page in the article, or may be graphic backgrounds on which you overlay text. Also discussed is how to insert videodisc frames or sequences into an article (this requires a special version of Hyperties™).

Graphics

The IBM PC supports several types of graphics. Unfortunately, not every computer can display all types; it depends upon the monitor and "graphic card" installed. In Hyperties™, we have attempted to shield the user from these hardware considerations. When you create a Hyperties™ database, you need not specify the type of graphic to be displayed. All you need to do is to tell the program where to insert a graphic into the article.

When a reader gets to a graphic used as an illustration, *The Browser* will retrieve it from the disk and examine it. It will determine if the graphic can be displayed on the computer being used to read the database. If the graphic can be displayed, *The Browser* will do so. If the hardware cannot display the graphic, *The Browser* will create a text page to substitute for the graphic. The text page informs the reader that a graphic was specified but it cannot be displayed on the computer being used. While this may be disappointing to the reader, it is not an error condition and does not inhibit use *The Browser.*

Background graphics are handled in the same way. If you specify that an article is to be displayed on a graphic background, *The Browser* will do so it if the computer on which the database is being run is capable of displaying it. If not, the background will be ignored and the text displayed without it.

Types of Graphics

Hyperties[tm] supports four types of graphics: TEXT, CGA, EGA, and VGA. Each is described below.

1. *Text graphics* are graphics made from the special "text characters" available on the IBM PC. They can be displayed on any PC, even one which has a text only (monochrome) capability. They are useful for tables and simple charts.

   You create a text graphic with any line editor or word processor which can produce a "clean ASCII" file (most can). When you save the text graphic, give it a file name of *xxxxxxx*.TXT (substituting any valid name for *xxxxxxx*).

2. *CGA Graphics* were the first type of graphic supported by IBM personal computers. They are relatively low resolution. They may be drawn in four colors (in which case they have a resolution of 320 x 200 dots) or in two colors (with a resolution of 640 x 200 dots).

   CGA graphics are generally not very exciting but they take up little disk space and can be displayed on any IBM PC with graphics capability. We have found the 2-color mode acceptable for simple graphic backgrounds and line drawings. An example

can be found in the Hyperties™ Tour in the article "CGA Backgrounds." If you are developing your databases for a computer with a monochrome monitor and a CGA card (typical on computers like the AT&T 6300), two color mode may be particularly useful.

The four color mode can produce acceptable color illustrations. Although you lose some resolution, you add considerable "zip." See the "newsletter" cover graphic in the Hyperties™ Tour for an example.

You may save a CGA graphic as an "uncompressed binary image", with a file name of *xxxxxxx*.PIC; this requires 16,391 bytes of disk space per graphic. If you create .PIC files, check the length. Some graphics editors assign the file extension .PIC to other formats; this will confuse *The Browser* and probably result in display problems.

You can also create CGA graphics with a program called PC PAINTBRUSH (a product of Z-Soft). PC PAINTBRUSH compresses the graphic so it requires less disk space. A file saved from PC PAINTBRUSH is named *xxxxxxx*.PCX (substitute any valid name for *xxxxxxx*). Because the .PCX format saves disk space, it is preferred.

3. *EGA Graphics* are higher resolution than CGA graphics. Because they can have more detail (640 x 350 dots) and up to 16 colors, they are generally more exciting than CGA graphics.

Hyperties[tm] supports EGA graphics created on PC PAINTBRUSH and saved with a filename xxxxxxx.PCX (where xxxxxxx is any valid name). If you have EGA graphics created with another graphics editor, there are programs available to convert them to .PCX format.

If you use EGA graphics, you should know that some computers were sold with EGA adapters equipped with less than 256K of memory. This is most likely true of IBM PC-AT computers equipped with IBM's own EGA card.

If an EGA adapter has less than 256K of memory on it, it cannot display 16 colors at a resolution of 640 x 350 dots, but (typically) can only display 4 simultaneous colors. Hyperties[tm] will check the memory available on an EGA adapter and will not attempt to display a 16 color EGA graphic on a computer without adequate display memory. This prevents the graphic image from being distorted.

4. *VGA Graphics* are standard on the PS/2 series of computer and may be displayed on earlier models (such as the PC XT and PC AT) if an appropriate card and monitor are installed.

   VGA graphics are higher resolution than EGA and can have either 16 colors (640 x 480 dots) or 256 colors (320 x 200 dots; this is the only multi-color VGA mode available on the PS/2 models 25 and 30).

Hyperties™ supports VGA graphics which have been created by PC PAINTBRUSH and saved in files named *xxxxxxx*.PCX (where *xxxxxxx* is any valid name).

Unfortunately, there is a problem with PC PAINTBRUSH's ability to store VGA graphics in 256 color mode. If you are developing graphics in VGA 256 color mode we suggest you save the palette separately ("SAVE PATTERNS" within PC PAINTBRUSH, naming the file *xxxxxx*.PAL where *xxxxxx* is the same as the name of the saved picture). Hyperties™ will recognize the need for the special palette file and load it to accompany the picture of the same name.

Including Graphics in an Article

To include a graphic in an article, simply insert the line:

@G=*xxxxxxx*

where you want the graphic. (You may also use lower case @g= *xxxxxxx*). For *xxxxxxx*, substitute the name of the graphic you want displayed. Note that you need not specify the file extension (.TXT, .PIC, or .PCX) with which the graphic is saved.

When the Browser encounters the command @G=, it will force a new page, go to the disk, retrieve the graphic, determine what type of graphic it is (text, CGA, EGA, or VGA) and display it. The usual options: NEXT PAGE, BACK PAGE, RETURN TO, and INDEX will be displayed at the bottom of the screen

although they may be abbreviated in CGA 4-color or VGA/MCGA 256-color resolutions (which can only display 40 columns of text).

If the graphic cannot be displayed because it is not on the disk, the disk is defective, or (most usually) the computer being used to browse the database does not have the ability to display graphics of the appropriate resolution, a text page explaining the problem will be constructed and displayed instead. The reader will be able to continue browsing without error.

Graphic Backgrounds

Graphic backgrounds allow you to mix text and graphics on a single page. There are many examples on the Hyperties[tm] Tour. Graphic backgrounds are used for three purposes:

1. To create interesting backgrounds for text. Examples are the "index card" and "yellow pad" used in the Hyperties[tm] Tour.

2. To create graphics in which text can be mixed with graphic elements. Example of this are the "angelfish" and "tetra" graphics in the Hyperties[tm] Tour.

3. To create graphics with selectable links. For example, maps with labels. An example of this is the "tropical fish" graphic ("Imbedded Illustrations" article) in the Hyperties[tm] Tour.

Graphic backgrounds may be created in any graphics mode, but it is advisable to only use modes which support 80 columns of text. This includes CGA 2-color mode but excludes CGA 4-color mode and VGA/MCGA 256-color mode.

You may specify a "standard" graphic background in the TIES.SYS file by including the line:

background=*xxxxxxxx*

where *xxxxxxxx* is the name of the graphic to be used as a background. No extension is required but one may be included.

Within an article, specify a graphic background by including the line:

@b=*xxxxxxxx*

at the beginning of the article. Do not attempt to specify more than one background per article and make certain that it is the first line. Otherwise you may get odd results in *The Browser*.

A background specified in the article will override the standard background. For example, you could set the "yellow pad" to be the standard background in the TIES.SYS file and then change it to the "index card" for a specific article.

If you have specified a standard background in the TIES.SYS file and wish to disable it (have no graphic background) in an individual article, you may insert at the beginning of that article the command:

@B=""

When you use a graphic background, *The Browser* must draw the text on the graphic, rather than using the hardware generated text characters. This slows the display down a bit. On most computers you will hardly notice it; but it may be noticeable on very slow computers.

Backgrounds Provided

There are two general purpose backgrounds used in the Hyperties[tm] Tour which you may wish to copy for your own use. They are in the directory labeled **\TOURPIX**. The graphics are:

YELLOPAD.PCX	a background which looks like a lined writing pad
NOTECARD.PCX	a background which looks like an index card

NOTE: Because the EGA adapter allows the 16 displayable colors to be selected from a total of 64, any EGA graphic can have a non-standard palette; this is true of these two backgrounds. The default highlight and cursor colors used by Hyperties[tm] will not work right because of these palette differences. If you want to make use of these graphics as backgrounds, set the color values in your TIES.SYS file to the values we use for the *Hyperties[tm] Tour*. See the TIES.SYS file in the TOUR directory.

Creating Your Own Backgrounds

You can create your own backgrounds by using PC Paintbrush. Backgrounds should be created in a mode which supports 80 column text (that is CGA 2-color, EGA, or VGA).

Controlling text colors in backgrounds can get a bit tricky and you need to understand how the display cards in your computer operate. When a background graphic is used, the palette of the background graphic and the colors selected for text mix and create different colors.

To aid in selecting colors to be used for text and highlighting on a specific background graphic, we have provided the **PALETTE** program on the *Author and Utilities* disk. Directions for using the program are in the *Addendum* at the back of this manual.

Videodisc

If you have an IBM Infowindow computer, you can display videodisc still frames and motion sequences. The procedure for inserting a videodisc display into an article is similar to that used to insert a graphic. At the point you want to display the videodisc sequence, add the line:

@V=*xxxxx-yyyyy*

where you substitute the frame number at which you want to begin for *xxxxx* and the frame number at which you want to end for *yyyyy*. When *The Browser* encounters the @V (or @v) command, it will force a new page and display the videodisc

sequence. The motion sequence will end at the last frame and the usual NEXT PAGE, BACK PAGE, RETURN TO, and INDEX options will be displayed, superimposed over the final frame which will be displayed as a still frame.

If you want to display a single videodisc frame ("freeze frame"), simply use the same number for *xxxx* and *yyyyy*.

To activate the "videodisc driver" you must specify the line

device=videodisc

in the TIES.SYS file. Alternatively you may start *The Browser* with the "switch" /DV:

browser *database* **/DV**

See the chapter on the TIES.SYS file for further information.

To use the IBM Infowindow, you must have a version of Hyperties[tm] which supports it. Contact Cognetics Corporation for information.

Converting Earlier Versions of Hyperties[tm]

Earlier versions of *Hyperties[tm]* required that graphic references be enclosed in carets (^). This is no longer the case. To convert articles prepared for use with earlier versions you may either:

1. Eliminate the carets entirely and imbed the graphic in the article with an **@g=** command

2. Replace the carets with tildes, for example: **^picture^** becomes **~picture~**. Then resolve the link to an article consisting of the single line

   @G = *xxxxxxx*

   This technique produces databases which work exactly like previous versions of *Hyperties™*.

Summary

Here is a summary of the main points covered in this chapter:

- You can insert graphics into an article.
- Graphics may be text files, CGA, EGA or VGA modes.
- The computer on which the database is browsed must have a graphics capability appropriate for the type of graphics contained in the database. If it does not, Hyperties™ will handle the problem gracefully but the user will not be able to view the graphics.
- Graphics are inserted into articles by including the command

  @G=*xxxxxxx*

  where *xxxxxxx* is replaced by the name you assign the graphic.

- Graphics are created using a graphics editor like PC PAINTBRUSH.

- Graphic backgrounds are inserted into articles by including the command

  @B=*xxxxxxxx*

  A "standard" graphics background can also be specified in the TIES.SYS file.

- If you have a special version of Hyperties[tm], you can insert videodisc sequences and freeze-frames into articles.

Chapter 6
Authoring Tutorial

In this chapter, you will gain experience with the Authoring Tool by creating your own Hyperties™ database.

First decide if you want to create the Hyperties™ database on a floppy disk or on your hard disk.

If you want to create the Hyperties™ database on a floppy disk, insert a formatted blank disk in drive B: and type the command:

MD B:\FISH

to create a directory to store the Hyperties™ database. Note that we do not recommend storing the database in the root directory since the number of files that can be stored in the root is limited by DOS.

If you want to create the Hyperties™ database on a hard disk, type the following command at the **C>** prompt

MD \FISH

to make a new directory for the database you are about to create.

Copying the Pictures

The database you will be creating, includes two pictures. Although Hyperties™ does not require that the pictures be present when you create the database (or even when it is run with *The Browser*), for convenience, we have provided the pictures for you. Your first step will be to copy the pictures onto your database.

The pictures are on the disk which contains the Hyperties[tm] Author. To copy them, insert the disk in drive A: and use the appropriate DOS copy command:

1. To copy the pictures onto a hard disk, type the command

   COPY A:\PICS*.PCX C:\FISH

2. To copy the pictures onto floppy disk B: type the command

   COPY A:\PICS*.PCX B:\FISH

Starting the Authoring System

Start *The Author* by typing in the command shown below:

1a. From a floppy disk, insert the disk with *The Author* in drive A: (and type " **A:**" if necessary to get the **A>** prompt).

1b. From a hard disk, type:

CD \HTIES

to activate the Hyperties[tm] directory.

2. To start *The Author*, type the command

AUTHOR *xxxxxxxxx*

replacing *xxxxxxxxx* with the name of the directory you created for your database (**\FISH** or **B:\FISH**).

The Index Screen

The *index screen*, shown in Figure 6.1 below, is your "home" screen. As there are no articles in the index yet, the article window is empty.

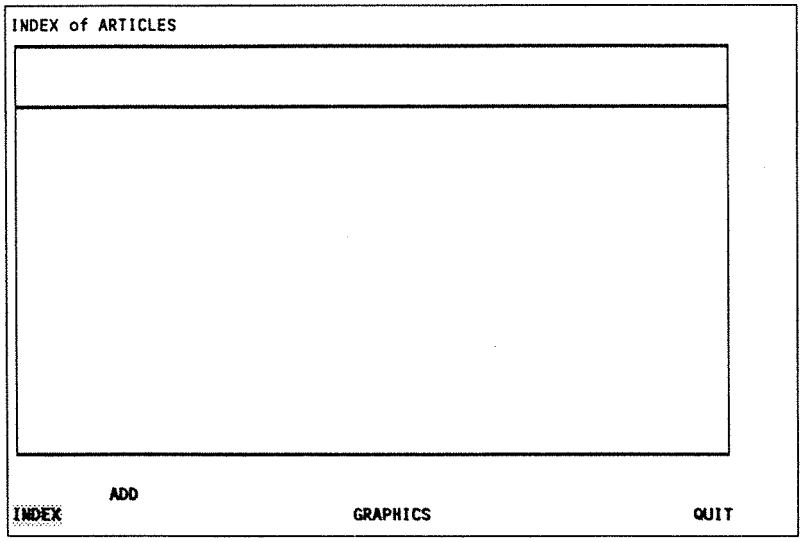


Figure 6.1
The Empty Article Index

Enter the first article into the index, by placing the selector bar on **ADD** and pressing ENTER. You will be asked for the name of the article you want to add. Type the name of the first article "introduction" then press ENTER.

By naming the article "introduction" it will automatically become the introductory article. Figure 6.2 below, shows the *index screen* after you have entered the first article into the index.

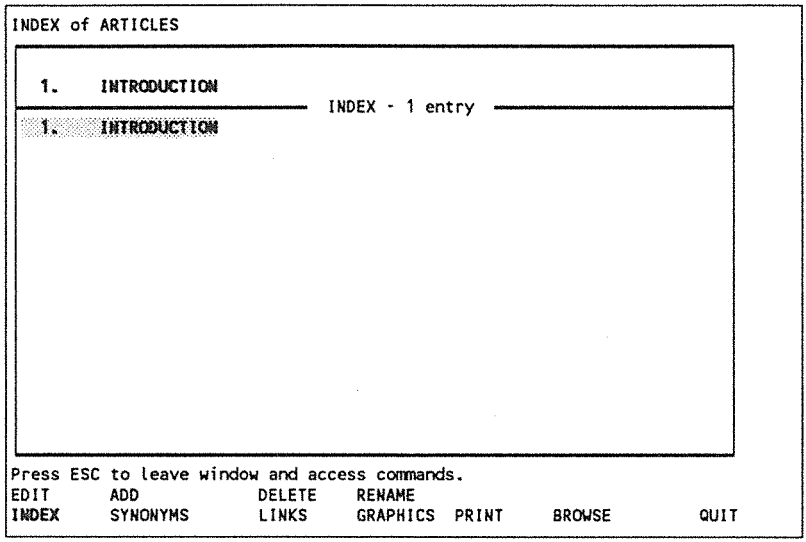


Figure 6.2
The First Article Added to the Index

If you wanted, you could add more articles to the index. Some Hyperties[tm] authors like to define most or all the articles at the beginning of the authoring session.

However, an alternative strategy is to create the article as soon as it is entered into the index. Press Esc to leave the window and place the selector bar on EDIT; then press ENTER to go to the *edit screen* shown in figure 6.3 below.

The *edit screen* has three windows. The first has the article title or **HEADING**. Hyperties[tm] has already filled this in for you, from the article index, since most of the time, you will want to display the article with the title the same as the index entry. If you wanted to change it, you could edit the **HEADING** window. If you want to, you can change the HEADING to "Fish are your Friends". If you do so, note that this will *not* change the way in which the article is entered into the index -- only the heading displayed when a reader looks at the article.

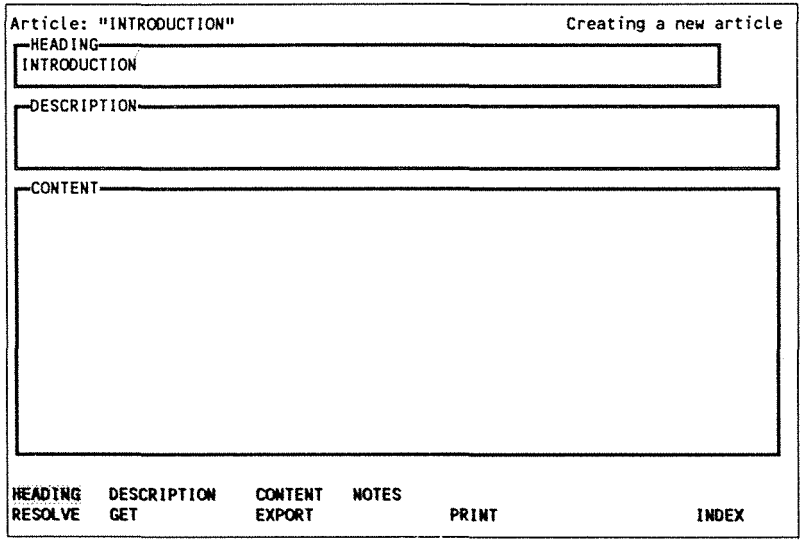


Figure 6.3
The Edit Screen

The second window is labeled **DESCRIPTION**. In this window you write a brief description or definition of the article. You will have two lines of text. When a reader selects the article, the definition will appear first. The reader may then decide whether to go on to the full content of the article (if you provide any).

The third window is labeled **CONTENT**. This is where you will enter the text of the article.

Enter the *brief description*. Since you are in the HEADING window, press ESC to leave it. The selector bar will move down to the bottom of the screen. Place it on the command **DESCRIPTION** and press ENTER. Type "An appreciation of our finny friends." Then press ESC to leave the window.

> **SPECIAL NOTE:** Under no circumstances should you type in a brief description which extends beyond the size of the editing window on this screen. If such a long description is allowed in your database, it is likely to give *The Browser* serious problems.

Next you will type-in the text of the article. Place the selector bar is on **CONTENT** and press ENTER. Type the text as shown

in Figure 6.4, below:

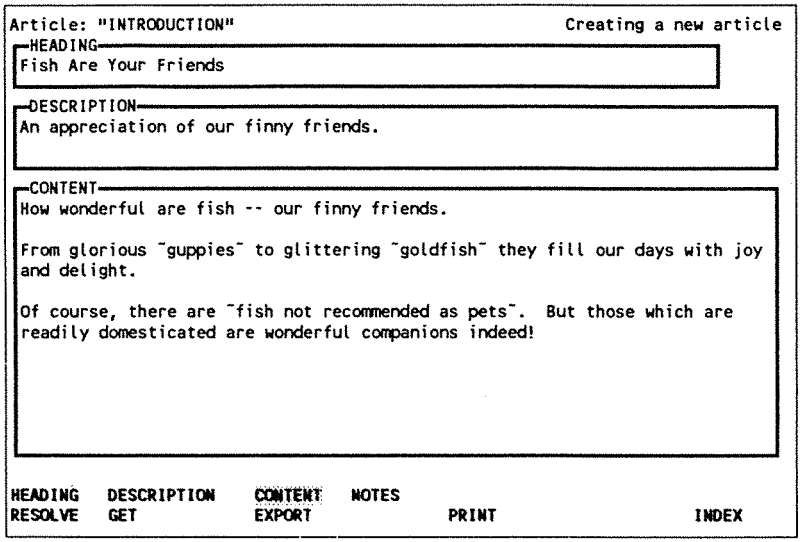


Figure 6.4
Entering Text

The second sentence contains two *links* or references: **guppies** and **goldfish**. Each of these words is enclosed in tildes (~) to inform Hyperties™ that they are links.

When you write the word "guppies", press the tilde key once and notice that two tildes (~~) appear on the screen. Type the word *guppies* and notice how the tildes expand to surround the word (~guppies~). Use the right arrow key to move the cursor past the second tilde. Type the word *goldfish* in the same way.

The feature of having two tildes appear when you press the tilde key is useful because it helps you avoid the error of mismatched tildes. However, sometimes you may want to enter a single tilde. (This happens, for example, when you want to go back and surround a word or phrase, already typed, with tildes). You may delete one of the tildes with the backspace or delete key.

Finally, enter the third paragraph. The third paragraph contains the phrase ~fish not recommended as pets~ which is also a link to a new article.

This article, therefore, has three words or phrases enclosed in tildes: ~guppies~, ~goldfish~, and ~fish not recommended as pets~. These three references are *links* to other articles (yet to be defined). When *The Browser* displays this article, the three terms in tildes will be highlighted and your reader may select them (providing, of course, that you define the new articles).

When you have entered all the text shown, press ESC to leave the **CONTENT** window. Now you are ready to resolve the three links in the article. Place the selector bar on **RESOLVE** and press ENTER to begin the resolution process.

Resolving the References

When you **RESOLVE** the article, Hyperties[tm] will examine the text and find the three links you entered. Every time Hyperties[tm] finds a link, it checks the index to see if the reference is already in it. If Hyperties[tm] finds the reference, it will link the two articles together automatically. But if the reference is not in the index, Hyperties[tm] will need your help to resolve it.

Recall that in this article, you specified three links: **guppies**, **goldfish**, and **fish not recommended as pets**. The first link

Hyperties™ finds is ~guppies~. It checks the index and finds that *guppies* is not yet in it. It will therefore ask for your assistance as shown in figure 6.5 below:

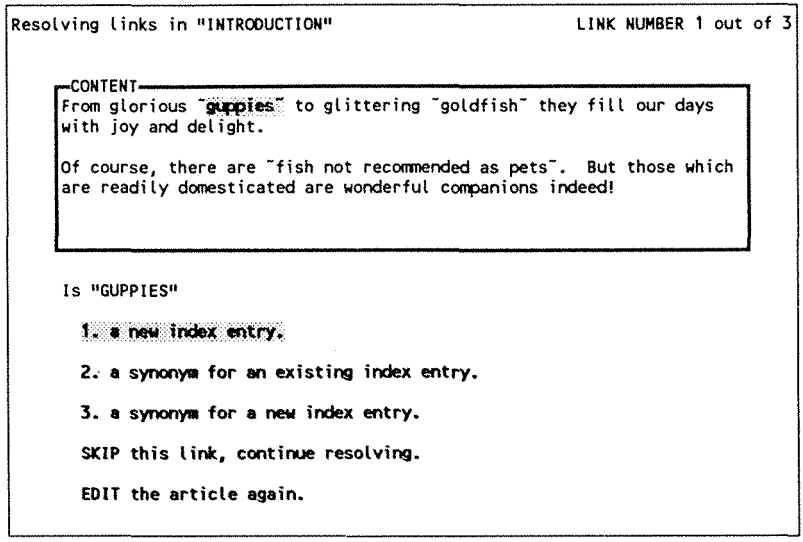


Figure 6.5
Resolving the First Reference

There are three possibilities: (1) *guppies* might refer to a new article not yet entered in the index (a new index entry), (2) *guppies* might be a synonym for an article already defined (a synonym for an existing index entry), or (3) *guppies* might be a synonym for an article you will define later (a synonym for a new index entry). In the present case, *guppies* refers to a new article to be input later, so it will be a new index entry.

Look at figure 6.5 above. The window at the top, labeled CONTENT, shows you the part of the text of the article containing the reference to be resolved. At the bottom of the screen you are shown the three possibilities:

Is "GUPPIES"

1. a new index entry
2. a synonym for an existing index entry
3. a synonym for a new index entry

Also included is a "skip" option:

SKIP this link, continue resolving.

and an option that allows you to go back and make a change to the article:

EDIT the article again.

The skip option is present in case you don't want to resolve this link. If you are confused about the first three options, re-read the paragraph above.

Since in this case, GUPPIES is a new index entry, select option 1. (a new index entry) and press ENTER.

Because you indicated that GUPPIES was a new entry, Hyperties[tm] adds it to the index. Now it asks you about the second unresolved entry ~goldfish~. This too will be a new article, so add it to the index.

Finally you are asked about the third unresolved reference ~fish not recommended as pets~. This too will be a new article, so add it to the index.

The processing of the article "introduction" is now complete. All three references have been resolved and are in the index. You will get a summary screen showing the three references and how they were resolved. A keypress will take you to the *index screen*.

Figure 6.6 shows the four entries in the index. Only one actually exists in the database, so far. The article which exists, has the characters ***t** next to it. The ***** indicates that a brief description has been entered and the **t** means that content has been entered.

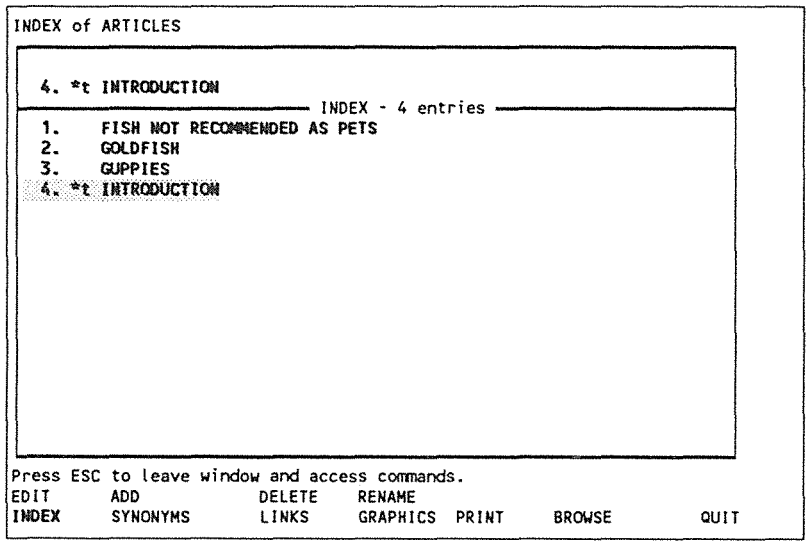


Figure 6.6
The Index After Resolving the First Article

Importing a File

You are now ready to enter the second article, "GUPPIES". You will *import* this article from a previously prepared file which we have supplied on the demonstration disk.

The article "GUPPIES" is already in the index, because it was referenced by the introduction. Make it the current article by placing the selector bar on it. Press ESC and select **EDIT** to get to the *edit screen* (**Note:** Here is a shortcut: you can perform any command directly by holding down the **Alt** key and pressing the first letter of the command. For example, **Alt-E** to EDIT).

Select **GET** on the *edit screen* to import the article. You will be asked for the name of the article to import. Make certain that the disk with the article is in drive A: and type

A:\TEXT\GUPPIES.TXT

and press ENTER.

You will see the light on drive A: come on as the file is loaded, and the article will be loaded.

Look at the content you have loaded. It uses a new feature; the line

@G=shark

is an instruction to *The Browser* to insert an illustration here.

Place the selector bar on the **RESOLVE** command and press ENTER to resolve the links (or simply press Alt-R).

There is one link in the article: **goldfish**. Since **goldfish** is already in the index; Hyperties™ will resolve the reference automatically.

Next, load the "Goldfish" article. Make "Goldfish" the current article and go to the *edit screen*. Load the file

A:\TEXT\GOLDFISH.TXT

Figure 6.8 below shows the text of the "goldfish" article. Note the use of the @T embedded formatting command to set *table mode*. Chapter 5 discusses this command in detail.

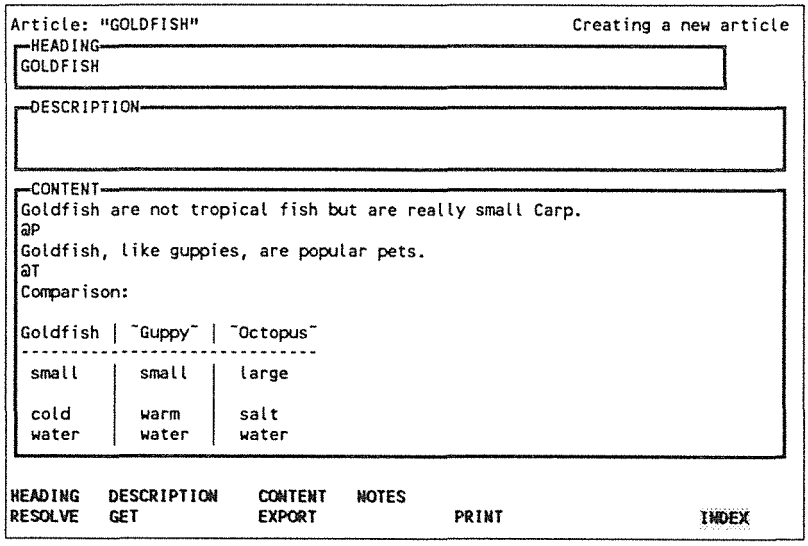


Figure 6.8
The Goldfish Article

Select **RESOLVE** to save and process the article. There is one reference, **guppy**, in the article. **Guppy** is a *synonym* for "guppies". If the reader selects **guppy** the article "GUPPIES" should be presented. Therefore when Hyperties™ asks you to resolve the reference, specify that it is a synonym for an existing entry as shown in figure 6.9 below:

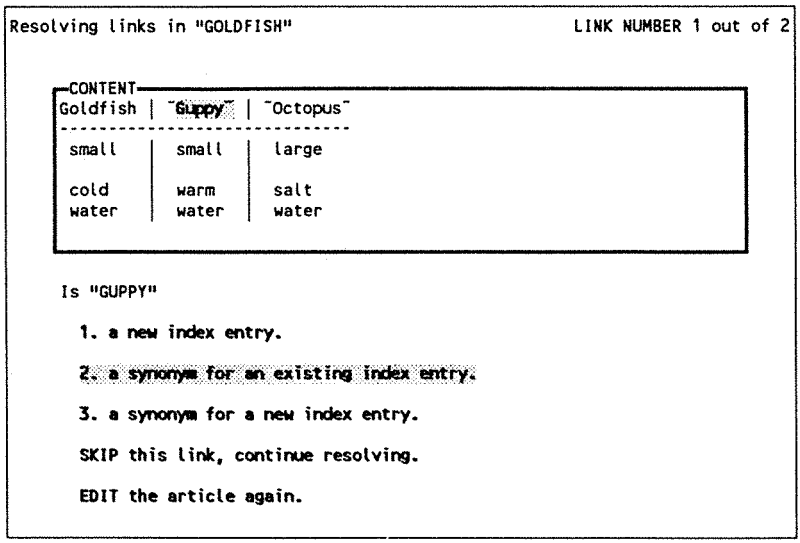


Figure 6.9
Specifying a Synonym to an Existing Article

Hyperties™ will ask you to specify the entry for which "guppy" is a synonym. Place the selector bar on "GUPPIES" and press ENTER.

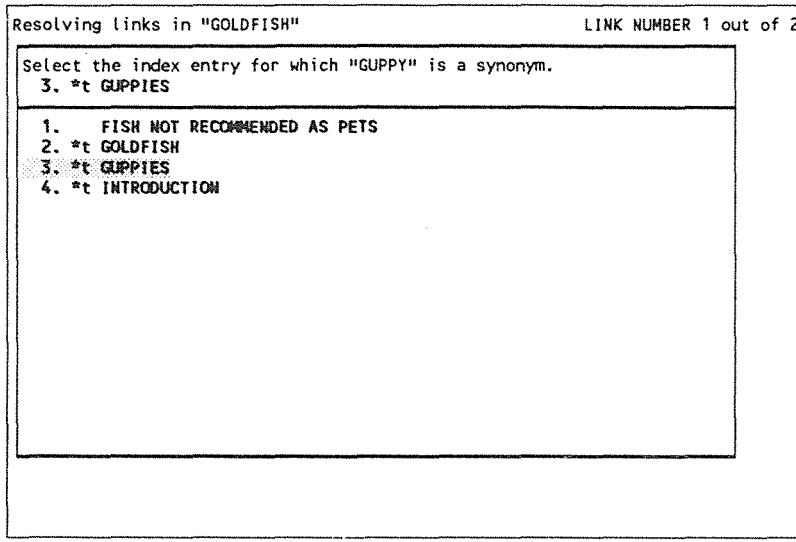
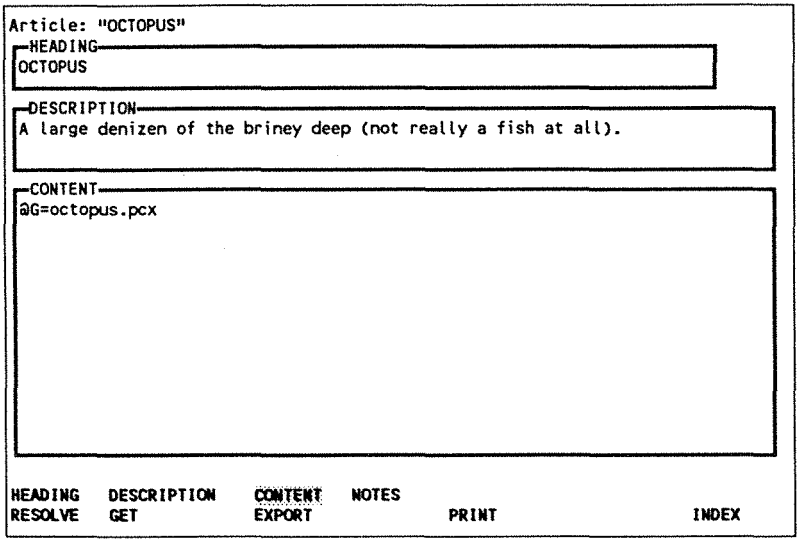


Figure 6.10
Resolving the Synonym

Now, load the article FISH NOT RECOMMENDED AS PETS. Load the text of the article from **A:\TEXT\NOT.TXT** on the demonstration disk. Use your imagination to create a brief description. Octopus is a new index entry.

Now create the article "octopus". Octopus is an article which consists only of a graphic. Type in the description and content as shown in figure 6.11 below. When you attempt to **RESOLVE** the article, you will find that all the references can be resolved automatically.



Create a similar "graphic only" article for "shark".

You will return to the *index screen* with your database complete. Note that the symbols ***t** appear next to each of the six articles in the index, indicating that a brief description and content exists for each.

Select **BROWSE** to see your database. When you quit *The Browser* (either from **EXTRA** or with the hidden command **CTRL-Q**) you will return to the *index screen.*

Place the selector bar on **QUIT** and press ENTER to leave *The Author*. Congratulations, your first Hyperbook is complete!

Creating the Search Index

Run the program *Update* to create an index for *The Browser* to use in searching. You must re-run this program every time you change the database. To run the *Update* program type:

UPDATE \FISH

if your data base is on the hard drive, or type

UPDATE B:\FISH

if it is on a floppy drive.

Browsing your Database

To browse the database and see your work, start *The Browser* with the command

BROWSER \FISH

or

BROWSER B:\FISH

depending upon whether you stored your database on a hard disk or a floppy disk. If you need additional instructions about starting or using *The Browser*, consult Chapter 3.

Note that once you create the search index files as described above, the **SEARCH** option become operative on *The Browser.*

Feel free to continue to experiment with the database you created. You can start *The Author* and make additional changes, then return to *The Browser* to view them as often as you would like.

You may also want to try *The Author* on a copy of *A Hyperties™ Tour.* You can modify the Tour and see how we created it.

Summary

Here is a summary of the material covered in this chapter:

- You may enter article titles directly into the index.
- When Hyperties[tm] encounters an unresolved reference, it will attempt to resolve it. If it cannot, it will ask if the reference is a new article or a synonym. Hyperties[tm] will enter new articles in the index.
- You may import articles from disk.
- You use the UPDATE program to create a search index.

Chapter 7
Authoring Strategies

This chapter will provide you with some suggestions as to how to use Hyperties™ effectively. As you work with Hyperties™, you will undoubtedly develop techniques of your own. We hope you will share them with us and, in turn, permit us to share them with other users in the Hyperties™ community. When you return your registration card, we will add your name to the Hyperties™ User's Group (HUG). Through HUG we will share ideas and help Hyperties™ evolve.

We have also set up an **electronic bulletin board**. Through this medium we hope to encourage sharing of databases and conferences. It will also provide you with a way of interacting with the Hyperties™ development team. You can access the bulletin board by calling (609) 275-6555. (You may communicate at 300, 1200 or 2400 baud. Set your communications program to 8 bits, 1 stop bit, no parity).

Planning the Database

Before you begin using the Hyperties™ Author we recommend that you decide what subjects your articles will be about and how the subjects of your articles will be interrelated. You will control the structure of your database by the ways in which you link articles together.

Designing the Introductory Article

The introductory (or lead) article is especially important. Since many of your readers will browse the introductory article first, this article should be composed so that it references as many key articles as possible. There are several strategies for composing the introductory article.

One strategy is to make the introductory article rich in references. In this strategy, you would make the introductory article a summary of the entire database. By scanning it, the reader can select one of many places to begin browsing.

A second strategy for the introductory article is to confine it to only a few key references. In this strategy, the idea would be to minimize the number of details which the reader must deal with and start him or her down an appropriate path. For example, suppose you were building a policy manual which had many detailed articles on specific policies. Rather than referencing many policies in the introductory article you could develop a more general approach such as the following:

> This database contains policies relating to: **permanent employees, temporary staff,** and **consultants**. In addition are policies relating to **security, non-disclosure,** and **dealing with the press**.

A third strategy for the introductory article is to design it as a high level index. You may find Hyperties' *table mode* to be useful here (see Chapter 4 for more information on table mode). Here is an example:

CORPORATE POLICIES

Permanent Staff
- **Hiring Permanent Staff**
- **Termination Procedures**
- **Benefits and Vacation Policies**

Temporary Staff
- **Approval Policy for Hiring Temps**
- **Approval Policy for Retaining Consultants**

This technique of using the introductory article as an index can be extended to other articles. For example, if the reader selected **Approval Policy for Hiring Temps** from the article above (s)he could be moved to a new article which provided a detailed "index" to relevant policies, for example:

APPROVAL POLICIES FOR HIRING TEMPS

Hiring Short-term Temporaries from Agencies
Establishing Qualifications of Temporary Staff
Hiring Independent Contractors

Using this technique you can create an extensive network of indexes. A particular article (policy in this case) could appear in many places within the indexes you create, so your readers can access it from many points. For example, an article on *Vacation Policies for Permanent Staff* could be highlighted under *vacations*, *benefits*, *permanent staff*, or any other relevant area.

Planning for Expansion

Do not hesitate to reference articles which you have not yet created. The Hyperties[tm] Browser will not highlight a reference to an article unless it exists. If you are writing an article and are discussing a topic which may eventually be the subject of its own article, you may put tildes around a word or phrase to setup a potential link. No harm is done if the referenced article is never written.

You have a choice, on the *resolve screen* (see Figure 6.9), to make the newly referenced term a *new index entry* or even to skip the reference, leaving the tildes in place for future use. If you let Hyperties[tm] create a new entry, you will be able to add content to the new article at a later time; at that moment, the link will be activated. If you decide to skip the new link when resolving, you will have to re-resolve this article later if you create an article to which the new reference should link.

However, if you do not insert tildes to indicate a reference, and you later decide to create an article on the would-be referenced subject, you have to re-edit the referencing article to insert the tildes as well as re-resolving it.

Writing Style

You may need to consider the most appropriate writing style to take full advantage of Hyperties[tm]. In general, it is best to keep articles short, so the reader need not deal with irrelevant text. Instead of expanding upon a subsidiary topic which is not the main subject of an article, you can merely mention it, delimiting the key word or phrase with tildes. Then you can make that topic the subject of its own article, or at least give the topic a definition which can be called up by the reader.

The same technique can be applied to details. Rather than including detailed information in an article you can simply reference it and create separate articles for it. This shields the reader from unnecessary details, but provides a path to them when the reader deems it relevant. This technique can be especially useful when the material contains case studies, experiments or many examples.

Creating Instructional Material

Hyperties™ can be used as a tool to reduce the difficulty of creating educational software and allow authors to focus on content and instructional design, rather than on technical factors. In Hyperties™, concepts and information can be entered and linked together. Developing courseware in such an environment is more like writing a book than writing a computer program. With Hyperties™, the development of courseware should become an instructional, rather than technical, endeavor.

For example, an introductory psychology module might contain the following text:

> The basic process in **behavioral psychology** was presumed to be **conditioning**. Two types of conditioning were extensively studied: **operant conditioning**, the more powerful form is most associated with **John Watson**; the less powerful paradigm of **respondent conditioning** is most frequently associated with the studies of **Ivan Pavlov**.

In the example above, the highlighted words and phrases in boldface indicate to the student that additional information is available on the topics: behavioral psychology, conditioning, operant conditioning, John Watson, respondent conditioning, and Ivan Pavlov. This additional information might be: a definition (for example, a definition of behavioral psychology), a new article (for example, an article on operant conditioning), an illustration (for example, a graph of conditioning and extinction), or a videodisc sequence (a brief biography of Ivan Pavlov).

Implicit within Hyperties[tm] is a cognitive model based on associative relationships. Articles in Hyperties[tm] explain concepts and the links which ties articles, illustrations and videodisc sequences together create relationships among concepts. A Hyperties[tm] database is therefore an associative network of concepts and examples at various levels. The power of this simple structure is enormous. Instructors can express the relationships among ideas by the manner in which articles are linked. Concepts can be expressed at multiple levels, with high level concepts linked to more specific concepts and specific concepts linked to examples.

Psychologists believe that human cognition is organized as associational networks of concepts. Therefore a Hyperties[tm] database uses a structure similar to the way people organize their cognitive structure. This suggests that courseware based on Hyperties[tm] will yield efficient learning.

Because a Hyperties[tm] database is organized according to the relationships inherent in the instructional material, it is excellent for helping students learn the material in an integrated, holistic fashion. One of the most difficult instructional tasks in any content area is conveying the systems and interrelationships which underlie the facts. Memorization of isolated facts leads to rote learning; integration of concepts and their relationship into the learner's cognitive structure leads to meaningful, useful learning.

Hyperties[tm] also creates courseware which is learner-controlled. Much computer-based instruction is based on a dialogue model in which the computer constantly prompts the student to respond to questions. This can be a powerful instructional model but is often implemented in a fashion which forces the student to accommodate to the pace and presentation units of the software. It is this factor which makes so much computer-based education unappealing to the student. In Hyperties, the student is in

control. Learning proceeds according to the pace and paths selected by the student. This not only provides the student with a sense of efficacy (essential to learning) but is an efficient way to present material. Students do not waste time on material they already know and they can pursue a topic of interest to any depth desired.

We do not suggest that Hyperties™ is ideal for all forms of instruction; rather, we suggest that it is excellent for certain types of instruction. In particular, Hyperties, we believe, will be extremely effective in the following areas:

- *familiarization* - situations in which a student is introduced to a new content area and needs to become familiar with the key concepts which underlie it and their relationships.

- *annotation* - situations in such areas as literature, poetry, art, law, and politics in which students read source material which may be heavily annotated. The annotations remain "hidden" behind the links and so do not interrupt the flow of primary material but are instantly available for reference.

- *dynamic glossaries* - glossaries prepared in Hyperties™ not only define terms but provide links to related concepts. This enables the student to more fully understand the key terms.

- *diagnosis and review* - coupled with objective test items, Hyperties™ becomes a powerful diagnostic and review technique for any achievement test. Students answer questions. If they are correct, the software moves on to the next item. If the student answers incorrectly, Hyperties™ presents an ex-

planation which serves as an entry to the database. Students can browse the database until they feel confident they understand the area and then return to the diagnostic test.

- *organizational information* - when there is a need to teach organizational relationships (for example, the structure of government) Hyperties[tm] can represent the relationships by linking informational articles together. Hyperties[tm] is also useful in presenting the facilities and services of an organization. For example, creating orientation courses for new college students.

Creating Reading Sequences

A final authoring strategy is the development of reading sequences (also called *tours*). By its nature, the articles in a Hyperties[tm] database can be randomly accessed. However, sometimes you may want to provide your reader with the ability to read the articles in some sequential fashion.

For example, suppose you had a set of material to present which you organized into five key articles. These articles contained the main ideas which you wanted all your readers exposed to. In addition, you might have written a number of more detailed articles, containing many details which expanded upon the key articles.

What you want the reader to do is:

(1) read all five key articles in sequence

(2) within a key article, use the Browser to explore the details of any concepts for which (s)he wants more information.

You can accomplish this in at least two ways. The first way would be to use the introductory article to list the five key articles:

> Please read the following key articles, in sequence:
>
> Chapter 1: **Buying Your Fishtank**
> Chapter 2: **Selecting Your Fish**
> Chapter 3: **Setting Up Your Tank**
> Chapter 4: **Feeding Your Fish**
> Chapter 5: **Care of Fish Babies**

The reader would select each article in turn. Within each article, the reader would browse related articles of interest and, and ultimately, return to the introductory article to select the next article.

The second technique is to create a link at the end of each article to the next article which you want the reader to see. For example, the first article *Buying Your Fishtank* might end as follows:

> Congratulations! Having followed the instructions in this article, you are now the proud owner of a fishtank. But what is a fishtank without fish? To move on to this exciting step, please read **Selecting Your Fish**, next.

This technique can be expanded to create several paths through the material, if you desire.

Summary

Here is a summary of the material presented in this chapter:

- Hyperties[tm] is an evolving product. We hope that you will work with us to share ideas and help it evolve.
- The introductory article is important because it is the entry point for many readers. There are a number of strategies for designing it.
- You may create references which refer to articles not yet in the database. When content is added to these articles, Hyperties[tm] will highlight the references.
- The ideal writing style for a Hyperties[tm] database may be different from that used in a conventional print medium because of Hyperties[tm] capability to hide details and discursions.
- Hyperties[tm] is a powerful system for creating instructional material because its basic structure is congruent with the way in which people store concepts.

Chapter 8
Installing Hyperties

This chapter provides information on installing Hyperties™ on your computer, on utilities provided for your convenience, and on technical information about Hyperties™.

Installation

This section will guide you through the process of unpacking and installing your Hyperties™ system. Step by step instructions are provided so new computer users will be able to install and run Hyperties™.

Hardware Required

To run Hyperties™, you need an IBM PC, PC-XT, PC-AT, PS/2, or a true compatible system, with a minimum of 256K of memory.

IBM computers are sold with four different types of displays: (1) *monochrome adapter*, (2) *color graphics adapter*, (3) *enhanced graphics adaptor*, and *VGA* (standard on the PS/2 series).

If your computer has a monochrome adapter *The Browser* will be able to display text but will not be able to display illustrations. The text will appear in light grey, green, or amber, depending upon the monitor attached to the computer. Hyperties™ will highlight links in boldface (extra bright) text. If there is an attempt to display an illustration, Hyperties™ will display a message instead, stating that the hardware is not capable of graphic display.

If your computer has a color graphics adapter (CGA) and color monitor, the Browser will be able to display both text and illustrations. Hyperties™ will highlight links by using color. However, the CGA text is not quite as sharp as that provided

by the monochrome adapter due to the fact that color monitors have less resolution than monochrome monitors.

If your computer has an enhanced graphics adapter (EGA) and high resolution color monitor, the Browser will be able to display both text and graphics. The text will be sharp and clear, and highlighted in color.

If your computer has VGA resolution, you will have the sharpest possible text and high resolution graphics with up to 256 colors.

Hyperties™ will automatically determine what kind of adapter and monitor it is running on and adjust itself accordingly. Therefore if you distribute copies of your Hyperties™ databases, they will run on any computer configuration. If the computer used to browse the database cannot display a particular graphic (because the graphic is higher resolution than the computer can display), the Browser will construct a text page with a message explaining the limitation.

You may run Hyperties™ on a computer with floppy disks or a hard disk. The most common configurations are:

- two floppy disks
- one floppy disk and a hard disk

Both *The Author* and *The Browser* will run on virtually any disk configuration; however, performance will be better on a hard disk.

License Agreement

Before you open the envelope containing the Hyperties™ disks, please read the license agreement printed at the beginning of this manual (pp. ix and x). We have tried to keep it reasonable and fair. Here are the key points:

- We are providing you with non-copy protected disks. We ask you to respect the years of hard labor which have gone into the development of Hyperties™ and not pirate our software.

- You may copy *The Browser* and distribute it with the databases you create, so long as you are not making more than the maximum number allowed in the License Agreement (see p. ix), or reselling *The Browser*. If you want to sell Hyperties™ databases commercially or need to distribute large numbers of copies of a database, let us know and we'll arrange a reasonable license agreement.

- We have done our professional best to provide you with a useful and bug-free system. We'll try to support your development efforts and correct any problems you find but we cannot accept liability for any bugs you might uncover. We want you to be happy with Hyperties™; if you're not, we'll refund your money within a reasonable time period.

If you find the terms of our license unacceptable, please return the package to us with the disk envelope unopened and we'll return your money.

Contents of the Disks

Inside the disk envelope you will find two floppy disks, as follows:

1. Disk 1 contains a copy of the Hyperties™ Browser and the demonstration database *A Hyperties™ Tour.*

2. Disk 2 contains the Hyperties™ Author and the text and graphics for the article used in the authoring tutorial (Chapter 6). This disk also contains the UPDATE utility used to create the signature files for searching and the PALETTE program for customizing colors in the TIES.SYS file.

If you received Hyperties™ on a 3.5 inch disk (such as used on the IBM PS/2 series) all the data is contained on a single disk.

Backing Up Your Disks

We recommend that the first thing you do is to make backup copies of the Hyperties™ disks and put the originals away in a safe place. You will need two floppy disks and labels (or a single 3.5 inch disk).

To copy the disks, perform the following steps:

1. Boot up your computer and make certain you have the DOS prompt (usually **A>** for a floppy-based system and **C>** for a hard disk).

2. *Follow this step if you have two floppy disk drives. (If you have only one floppy drive, follow the procedure in step 3).* If you do not have a hard disk

make certain that the DOS diskette is in drive A. Then type:

DISKCOPY A: B:

The computer will respond with:

Insert SOURCE diskette in drive A:
Insert TARGET diskette in drive B:
Press any key when ready . . .

Insert the Hyperties™ diskette in drive A: and the blank diskette in drive B:, then press **ENTER**.

In a minute or so you will see the question:

Copy another (Y/N)?

To copy another disk, type a "y" and press **ENTER** to copy the next disk. Repeat this process until all disks are copied.

WARNING: BE CERTAIN THAT THE DISKS YOU ARE COPYING ONTO ARE YOUR BLANK DISKS AND NOT THE Hyperties™ DISKS YOU RECEIVED!

3. *Follow this step if you have one floppy disk drive. (If you have two floppy drives, follow the procedure in step 2).* If you do not have a hard disk, make certain that the DOS diskette is in drive A. Then type:

DISKCOPY A: A:

The computer will respond with:

Insert SOURCE diskette in drive A:
Press any key when ready . . .

Insert the Hyperties™ diskette in drive a: then press **ENTER**. In a short time, the computer will instruct you to remove the Hyperties™ diskette (SOURCE diskette) from drive A: and replace it with the blank diskette (TARGET diskette). Do so, and then press **ENTER** . You will need to swap diskettes several times to complete the copy process.

When the copy is complete, you will see the question:

Copy another (Y/N)?

You may type a "y" and press **ENTER** to copy the other disk. Repeat this process until both disks are copied.

WARNING: BE CERTAIN THAT THE DISKS YOU ARE COPYING ONTO ARE YOUR BLANK DISKS AND NOT THE Hyperties™ DISKS YOU RECEIVED!

You should now have copies of the Hyperties™ disks you received. Label them carefully and put the originals away in a safe place. If you plan to run Hyperties™ from floppy disks you are ready. If, however, you want to install Hyperties™ on a hard disk (which we recommend if possible), follow the instructions in the next section.

Installing Hyperties™ on a Hard Disk

Installing Hyperties™ on a hard disk will make it more convenient to use. Installation is simple; just follow the steps below:

1. Insert the disk labelled **Browser and Tour** in a drive (typically drive A:) and make it the current drive by typing the appropriate DOS command (usually **A:**) and pressing ENTER.

2. Type:

   INSTALL A: C:

   to start the install procedure. If necessary, substitute for "A:" the letter of the drive containing the floppy; substitute for "C:" the letter of your hard drive. (For example: INSTALL B: D:).

3. When prompted, remove the **Browser and Tour** disk and replace it with the disk marked **Author and Utilities**. This is not necessary if you are installing from a 3.5 inch floppy.

Chapter 9
The TIES.SYS File

This chapter describes the **TIES.SYS** file which allows you to customize the way *The Browser* operates. Also described is the **UPDATE** program which is used to create the signature files (index) used for searching the database, and the **PIC2PCX** program utility which converts pictures from previous versions of Hyperties™.

The TIES.SYS file

Hyperties™ Version 2.3 allows you to specify particular files to be used as cover page and help page. In addition, you can instruct *The Browser* to use a mouse or touchscreen for input, and can set the colors used for highlighting and selecting links.

All of these options can be included in a file called **TIES.SYS**, located in the same directory as the particular Hyperties™ database. *The Browser* will use any specifications listed in the TIES.SYS file to override the usual settings for a file or color.

To create or modify the TIES.SYS file, use any ASCII editor such as EDLIN which came with your version of DOS. The specifications the Browser recognizes are:

HIGHLIGHT-COLOR = *n*
CURSOR-COLOR = *n*
SELECTION-COLOR = *n*
TEXT-COLOR = *n*
COVER = *filename*
HELP = *filename*
DEVICE = *x*
GRAPHICS = *directory*
BACKGROUND = *filename*
DESCRIPTION = *switch*

Each specification should be on a line by itself.

Setting Colors

You may set basic text color, the highlight color (the color which is used to display links), the cursor color (which is used for the highlight bar) and the selection color (which is the color a link becomes when you press ENTER). To change the colors, add the following lines to TIES.SYS:

TEXT-COLOR = *n*
HIGHLIGHT-COLOR = *n*
CURSOR-COLOR = *n*
SELECTION-COLOR = *n*

Replace *n* by the number of the color as specified in your computer's manual; this number is the sum of the foreground color number (0 to 15) and the background color (0 to 7) multiplied by 16 (16, 32, 48, etc.). Because this can be difficult, we have provided a program called **PALETTE** which you can run to select the colors and get the numbers.

The standard IBM colors are:

0 = Black;	1 = Blue;	2 = Green;	3 = Cyan;
4 = Red;	5 = Magenta;	6 = Brown;	7 = Lt Gray;
8 = Dk Grey;	9 = Lt Blue;	10 = Lt Grn;	11 = Lt Cyan;
12 = Lt Red;	13 = Lt Mag;	14 = Yellow;	15 = White.

See the special **Addendum**, at the end of this manual, which gives detailed instructions on the use of the PALETTE program.

Problems with Monochrome Displays

Some computers are equipped with monochrome monitors but use a color graphics adaptor. On these computers, the colors are transformed into shades of green, amber, or grey. Occasionally you may find one of these computers on which the highlighting does not have enough contrast to distinguish it from normal text.

On such a computer you should either set the mode to text (consult your manual) or alter the highlight color to one which shows up well. (The PALETTE program will help you set the colors appropriately.)

Specifying a Cover

Every book needs a cover. You may include one in your Hyperties™ database if you wish. If you don't specify a name for the cover, *The Browser* will look for a file called **COVER.***xxx* (like COVER.PCX or COVER.TXT). You may change the name of the cover by inserting the line:

COVER = *filename*

in the TIES.SYS file. Replace *filename* with the name of the text file or graphic you want to use.

The Help Page

When the reader presses a **?** or **CTRL-H**, the help page is displayed. A standard help page called **HELP.TXT** is provided with the system and you may copy it on to your own databases if you wish.

To change the name of the help file, add the line:

HELP = *filename*

to the TIES.SYS file. Replace *filename* with the name of the file you want to use.

It is likely that we will replace this help function with a more sophisticated one in the next version of Hyperties[tm].

Specifying a Mouse or Touchscreen

Adding the line:

DEVICE = *x*

to the TIES.SYS file lets you specify the input device(s) to be used. If no device is specified, *The Browser* assumes the keyboard will be used.

Replace *x* by a letter (or group of letters) specifying the input device to be used. The following codes are supported in the standard version of *The Browser*:

M (mouse)
K (keyboard)
N (no keyboard - keyboard is disabled)

(Special versions of *The Browser* are available which support **T** for touchscreen and **V** for videodisc.)

SPECIAL NOTE FOR MOUSE USERS: You will need to make sure that the DRIVER program for your Mouse is loaded into the computer before running Hyperties[tm]. This program should have been supplied with the Mouse when purchased, and is usually called something like PCMOUSE or MOUSE.

We have tested *The Browser* with the Microsoft mouse driver only.

Setting a Graphic Directory

You may mix graphics in the same directory as your articles or you may create a separate directory for them. If you create a separate directory, add the line:

GRAPHICS = *directory*

to the TIES.SYS file. Replace *directory* with the name of the directory you are using. Use a full pathname if needed. For example:

GRAPHICS = c:\MYDATA\PIX

Graphic Backgrounds

You may specify a standard graphic background in the TIES.SYS file with the line:

BACKGROUND = *filename*

Replace *filename* with the name of the graphic (an extension is optional). If you specify a graphic background in the TIES.SYS file, it will be used as a standard background in all articles except those which specify a background by including the line:

@b=xxxxxxxx

If you do not specify a background in the TIES.SYS file, none will be used, except where specified within an article.

Note that *The Browser* will not display a graphic background unless the computer on which it is run has an appropriate graphics card.

Turning Brief Descriptions On or Off

You may change the way in which Hyperties[tm] displays BRIEF DESCRIPTIONS by including in the TIES.SYS file the line:

DESCRIPTION = *switch*

where *switch* is one of the following words:

always **present** **never**

The standard (default) option is to **always** display the brief description. This means that when a link is selected the reader will have to consciously select the "GO TO COMPLETE ARTICLE" option, even if there is no brief description displayed.

The second option, **present**, displays a brief description if it exists; otherwise the reader is taken directly to the linked article.

The **never** option ignores all brief descriptions and always jumps immediately to the linked information. This is particularly useful in public-access or exhibit situations with touchscreen input.

UPDATE: Updating the Search Facility

After you have finished editing a Hyperties[tm] database, you will need to run the UPDATE program before the SEARCH function will work in *The Browser*. To use UPDATE, you simply type:

UPDATE *directory* **[ENTER]**

UPDATE analyzes all the files in the specified directory and creates the two Signature Files (indexes) used by SEARCH, placing them directly in the database directory.

Converting Graphics

Specifically for those who purchased earlier versions of Hyperties™, we have provided a utility program to convert graphics from PC Paint's .PIC format to PC Paintbrush's .PCX format. We call this utility **PIC2PCX**.

> **NOTE:** This is NOT a general-purpose utility for converting all PC Paint pictures. It ONLY works on 16,391-byte long, non-compressed CGA graphic files. This type of file is one which is loadable into BASIC with the BLOAD command, and is created by PC Paint's CAPTURE program if Capture has been started with a "/b" option.

To use PIC2PCX, you must specify the filename of the graphic you wish to convert. The program will either take the name as part of the calling procedure, or it will prompt you for a name after it starts. So you may type either:

PIC2PCX *filename*.**PIC <ENTER>**

Or:

PIC2PCX <ENTER>

PIC2PCX will load the specified graphic and then display this menu on the top line:

<P>allete <B>ackground <C>onvert <ESC>

You may use the P or B keys to change the color scheme of the displayed picture - repeated presses of the P key will cycle

through the available color palettes (4 standard sets), and repeated presses of the B key will display each of the 16 different background colors. When the colors are to your liking, you may save the graphic in .PCX format by pressing the C key. If the original picture was called "SUNSET.PIC", the program will create a new file called "SUNSET.PCX" in the same directory.

Distributing Copies of a Database

Once you have created a "Hyperbook" you will probably need to create copies for distribution to your own clientele of readers. This section describes the organization of a Hyperties[tm] database distribution diskette.

The root directory of the diskette should have a copy of BROWSER.EXE and probably a BATCH file that will automate the execution of *The Browser*. This BATCH file can be created with any simple text editing program, or by typing in from the keyboard:

COPY CON *filename*.**BAT <ENTER>**
BROWSER *directory* **<ENTER>**
<CTRL>-Z <ENTER>

In this example, the BATCH file is called *filename*.**BAT** and the text of the file is "BROWSER *directory*". A BATCH file is simply a way to substitute one word for a multiword command or series of commands. The name of the BATCH file can be used in place of typing-in the commands.

Your diskettes should then have a sub-directory which contains the database files. This directory should have copies of all the files in your working database directory. One way to copy all

these files from one disk to another is with the COPY command and *'s for "wildcards":

COPY A:*directory.* B:*directory***

The first drive\directory combination is the SOURCE, and the second is the DESTINATION.

Your complete database should include files with these names:

INDEX.TIE
REFER.TIE (Not required once editing is finished)
TIES.SYS (Optional)
SIG.P (If you have run the UPDATE program for searching)
SIG.S "
_____.FIL (Text of an article)
.....
.....
etc., one .FIL file for each article in the database.

Your database will also contain whatever .TXT or .PCX files you are using. These may optionally be placed in a second sub-directory, as long as that directory is named in a "GRAPHICS=..." line in the TIES.SYS file.

Chapter 10
Glossary/Index

This chapter is a glossary which defines various technical words used in this manual. It also contains an index to the manual.

Glossary of Terms Used in This Manual

actions

Actions are commands which you use to instruct the Hyperties™ Author and Browser to perform. Actions are always **highlighted**; you select and action by placing the *selector bar* on it and pressing ENTER.

article index

The index contains an entry for each article in the Hyperties™ database. It does not display any synonyms for the article entry. The articles are displayed in alphabetic order.

article screen

The article screen or edit screen is the screen in the Author Tool on which you create or edit articles. It contains three windows for the *heading*, the *brief description*, and the *text* of the article.

ASCII

American Standard Code for Information Interchange. Article text which is imported from another word processor must be ASCII files. When the phrase "clean ASCII" is used, it means that the editor or word processor you are using should save the text as ASCII characters and should not embed specialized codes of its own in the text. Files to be "imported" and text graphics, should be saved as clean ASCII files.

Author, the

The software program which is used to create and modify a Hyperties[tm] database. *The Author* contains a built-in word processor and the logic to create the various indexes used by the Browser.

Browser

The Browser is the software program used to read a Hyperties[tm] database. It is normally distributed with the database.

built-in word processor

The Author contains a built in word processor (editor) which provides a convenient way to enter text. If you prefer, you may use your own editor and import the files into the database.

CGA

Color Graphics Adapter. One of the graphics boards which can be used with an IBM PC. The CGA produces low resolution graphics with limited colors but is widely available. Another advantage of CGA graphics is that the amount of disk space required to store them is low. See also: EGA and VGA.

cognitive structure

A psychological term referring to the way in which people organize and store information.

color monitor

The color monitor used on an IBM PC or compatible must be matched to the graphics card used (CGA, EGA, or VGA).

commands

Instructions to *The Author* or *The Browser* to perform some action. Commands are also called actions. In most cases, commands are located at the bottom of the screen.

courseware

Computer presented instructional material.

current article

The article on which you are currently working in *The Author.*

cursor

In the Hyperties™ system, the cursor is also called the selector bar. The selector bar is always on the screen and is moved from one highlighted term to another by means of the arrow keys, a mouse, or a touchscreen. The bar can also be moved by pressing the first letter of the term to which you want to move it. Each press of the letter will cause the bar to jump to the next term which begins with the given letter. When the highlight bar is on the desired term, pressing ENTER (or "clicking" the mouse) instructs the computer to act on it. (You do not have to press ENTER when using a touchscreen).

database

A Hyperties[tm] database is a network of pictures, articles and definitions, linked together by highlighted terms or *references*. The database is sometimes referred to as a hyperbook because of its similarity in structure (cover, preface, table of contents, index, articles and illustrations) to a conventional book, translated into hypertext. The network or articles and links is also known as a *web*.

default

Default values are those assumed by Hyperties[tm] in the absence of other specifications. Examples include the colors to use for text display and where to look for graphics. To override the default values, you use the TIES.SYS file. Consult the chapter on TIES.SYS for details.

edit

In computer terms, editing refers to any modifications you make to a text file. A program such as a word-processor used to create and modify text is known as an *editor* or *text editor*. A program used to create or modify graphics is known as a *graphics editor*.

EGA

Enhanced Graphics Adapter. A circuit card which can be installed on IBM personal computers. When paired with an appropriate color monitor, the EGA adapter produces easy to read text in color and medium resolution graphics.

embedded formatting commands

Embedded formatting commands are inserted in the text of an article and instruct *The Browser* how to present the article on the computer screen. They are also used to specify the name of a graphic or the frames on a videodisc to be displayed. Embedded formatting commands always begin with the character @.

highlight

In Hyperties™, every action and every *link* (reference to additional information) is highlighted. On a monochrome monitor, highlighted text appears in boldface. On a color monitor, the highlighted text appears in a color different from the rest of the text. The standard highlight colors may be changed by specifying an appropriate line in the TIES.SYS file. A utility program called PALETTE is provided to facilitate this.

The *selector bar* may be placed on any highlighted item.

hyperbook

A hyperbook is the hypertext analogy to a conventional book. Unlike a conventional book which is normally read sequentially, a hyperbook allows the reader to access its information selectively in many different ways.

hypertext

A collection of techniques for storing and retrieving information on a computer. Hypertext is usually distinguished from linear techniques in which information is retrieved "from beginning to end" and from "brute force"

search techniques which simply locate all occurrences of a given piece of information without regard to its context.

Hyperties™ Tour

A demonstration disk which illustrates some of Hyperties™ capabilities.

illustration

Illustrations in the Hyperties™ database are computer graphics developed with a graphics editor such as PC PAINTBRUSH. A special version of Hyperties™ is available which allows videodisc sequences to be used as illustrations.

import

To load a text file created on a word processor into the Hyperties™ database. The text file must be "clean ASCII", meaning that it does not contain embedded formatting codes for the word processor on which it was created. (It may, of course contain Hyperties™ embedded formatting commands).

index entry

The entry for an article in the index. Often the index entry is the same as the article's *heading* which is displayed on the top of the screen when the reader requests the article. However, you may change the heading if you wish.

index window

The index window displays all the index entries in alphabetical order. You may enter this window and place the selector bar on any entry. Then press ENTER and your choice will appear in the index entry window.

introductory article

The introductory or lead article is the first article encountered by the reader. To get to this article, your reader selects INTRODUCTION from the preface (if it exists) or from the title page. There is a discussion on techniques of writing the introductory article in the chapter on Authoring Strategies.

lead article

see introductory article.

license agreement

The license agreement defines the terms on which you may copy and distribute Hyperties™.

link

A link is a reference to an article. Links are specified by enclosing them in tildes (~) when the article is created or modified. When the article is saved by the author, the links are *resolved* to point to the articles to which they refer. When links are displayed by the Browser, they are highlighted.

margins

Unless you change the margins with an embedded formatting command, the Hyperties[tm] Browser will use a right margin of 5 and a left margin of 75 when it displays the article.

monochrome adapter

A monochrome adapter card, connected to a monochrome (usually green or amber) monitor, provides clear text on an IBM PC. It cannot, however, display graphics.

notes

Hyperties[tm] allows you to record notes associated with an article. These notes are not displayed to your reader but can be used by you to record any information you desire.

reference

A reference is a highlighted term or phrase which appears in the text of an article. It is a link to additional information. When the selector bar is placed on a reference and the ENTER key is pressed, Hyperties[tm] displays additional information about it.

scroll

A technique of moving text up and down on the computer's display so you can see it all. Scrolling is necessary when there is more text than can fit on the screen or window.

Hyperties[tm] using scrolling in some places in *The Author.* However when presenting articles, *The Browser* uses *paging*

(NEXT PAGE and BACK PAGE) because the research has shown that readers find it easier.

selector bar

The selector bar or cursor can be positioned on any highlighted word or phrase. To move the selector bar, press the arrow keys, use a mouse, type the first letter of the highlighted term you want (several presses may be needed if more than one term begins with the same letter). Once you have selected a highlighted term (action) press ENTER to perform it. The selector bar is used both with *The Author* and *The Browser.*

synonym

One or more synonyms may be defined for any article. Synonyms allow you to refer to an article by different names. The use of synonyms can improve writing style.

table mode

Table mode is turned on and off by the embedded formatting command @T. In table mode, the Browser displays text exactly as it was entered. This makes it easy to display tabular information, and simple charts and graphs.

This type of information may also be created as a text file (with the extension .TXT) and displayed as a graphic using the @G= embedded format command.

TIES.SYS

TIES.SYS is a file which you use to specify options for *The Browser.* If you do not include a TIES.SYS file in your

database, *The Browser* will use *default* or standard values. See the chapter on the TIES.SYS file for details.

tildes

Tildes (~) surrounding a word or phrase instruct Hyperties[tm] to treat the word or phrase as a link to another article. If the article referenced exists in the Hyperties[tm] database, the link will be highlighted by *The Browser.*

touchscreen

Hyperties[tm] can operate with monitors which have a touchscreen. When a touchscreen is used, the reader simply touches the highlighted term with his or her finger. Currently, version of Hyperties[tm] are available which support the IBM Infowindow touchscreen and the Microtouch touchscreen. Contact Cognetics Corporation for details.

videodisc

Hyperties[tm] version 2.3 supports videodisc on the IBM Infowindow and with other videodisc control cards. Contact Cognetics Corporation for details.

windows

Windows are screen areas which contain text or lists of highlighted items. Most windows have labels in the upper left corner. To enter a window, place the selector bar on the COMMAND (usually at the bottom of the screen) which matches the label, and press ENTER.

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The Palette Utility - an Overview

The Palette program facilitates customizing the text colors used in your Hyperties[tm] database. You may set the color of:

normal text
highlighted text (links & commands)
selector bar

And, although customization of the following will only be available in Version 2.4 and beyond, you can look at different colors for:

screen background color
screen border color

Palette allows you to experiment with different color schemes until you find one that suits you, and it can automatically modify the TIES.SYS file to set those colors for an entire database.

If you are working with background graphics, Palette allows you to load a specific background graphic and choose text colors that look best with it.

Starting Palette

To start Palette, make sure that the current directory is your Hyperties[tm] directory, (CD \HTIES for hard disks, or insert the working copy of the *Author & Utilities* disk in drive A:) and type:

PALETTE *directory* [ENTER]

where *directory* is the location of the database with which you are working. If you do not specify a directory, Palette will work with the current directory.

The Palette Main Menu

There are two slightly different main menus, depending on whether or not you are currently viewing a background graphic. Palette starts in text mode with the menu in Figure A.1. When you are viewing a background graphic, the menu in Figure A.2 is displayed.

```
Would you like to set:

--> Text Color
--> Highlight Color
--> Selector Bar Color

--> Screen Border Color
--> Screen Background Color

--> Background Graphic
--> DONE
```

Figure A.1
The Main Menu (text mode)

On either menu, you select a choice in the same way commands are chosen within Hyperties™: move the selector bar by using the arrow keys and then press ENTER.

When you select any of the color change options, a window will open on the screen with samples of all available choices. The current color will be marked by a pair of arrows; these pointers can be moved from sample to sample using the keyboard cursor-control arrows. As the pointers move, the entire screen will change to show you how the indicated color will affect a Hyperties™ page. Press ENTER to select a particular color and move back to the main menu.

Credits

Hyperties[tm] was developed under the leadership of **Professor Ben Shneiderman**, Head of the Human/Computer Interaction Laboratory at the University of Maryland. Dr. Shneiderman developed the concept of Hyperties[tm], supervised its development, and coordinated a research program to evaluate its design.

Other participants in the design were **Dr. Janis Morariu** Assistant Director of the Center for Instructional Design and Evaluation, at the University of Maryland; **Dan Ostroff**, formerly a graduate student at the University of Maryland and currently a systems designer at the Museum of Jewish Heritage in New York; and **Dr. Charles B. Kreitzberg**, President of Cognetics Corporation.

Professor David Altshuler and **Anna Cohn**, encouraged, supported, and helped define the original concept when they were both on the staff of the U.S. Holocaust Memorial Museum and Education Center in 1983. Dr. Altshuler has continued to support the development of Hyperties in his new role as Director of the Museum of Jewish Heritage in New York.

Many individuals contributed to the software development. At the University of Maryland: **Jacob Lifshitz, Raymond Lee, Yuri Gawdiak, Richard Potter, Bill Weiland, Susan Flynn, Marie Seifring, Yoram Kochavy** and **Dr. Catherine Plaisant-Schwen**. At Cognetics Corporation: **Paul Hoffman** and **James Terry**.

The user's manual was written by Charles B. Kreitzberg with support from Paul Hoffman.

You may send correspondence regarding research and design issues to:

The Hyperties[tm] Project, Human/Computer Interaction Laboratory, The Center for Automation Research, The University of Maryland, College Park, MD 20742.

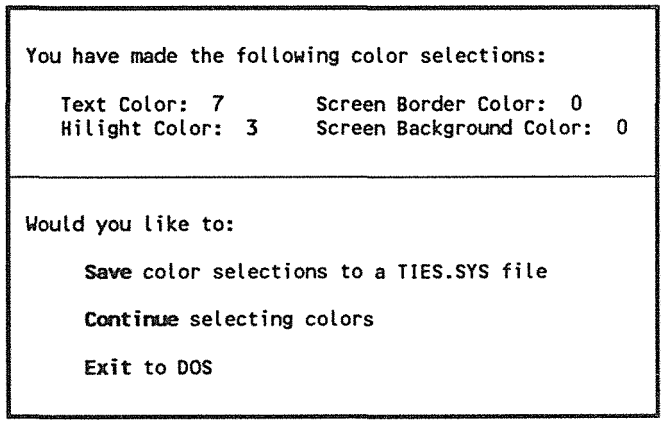


Figure A.3

You may either:

Save your color selections to a TIES.SYS file (with complete freedom to pick the directory);

Continue selecting colors (go back to the Main Menu); or

Exit the Palette program, returning to DOS.

Notice that the screen informs you what your color selections were; this is very helpful if you are going to set the colors within a specific article (using the @CT= or @CH= embedded commands). Simply mark down the color numbers and then embed the appropriate commands at the beginning of the particular article (see p. 73 for information about embedded color commands).